Employment Persistence after Raising the Pensionable Age: Evidence from Japan[1]

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Abstract

This paper examines how raising the pension eligibility age affects retirement behavior in Japan. Using administrative tax data, we analyze a reform that increased the eligibility age for pension benefits from 60 to 61 for certain cohorts. Leveraging a difference-in-differences approach, we find that affected cohorts increased their employment at age 60 by approximately 5 percentage points relative to unaffected cohorts. Moreover, the employment effects persisted beyond the new eligibility threshold: affected cohorts remained more likely to work between ages 62 and 64, even after becoming eligible for pension benefits. As a result, the reform induced roughly 60 percent of individuals who otherwise would have exited labor force to work approximately three additional years on average. This persistent effect is unlikely to be driven by income or wealth effects. Instead, it reflects two key features of Japan's labor market institutions at older ages: (1) workers are mandated to resign new contracts at age 60 to continue employment, often involving substantial changes in work arrangements; and (2) strong employment protection rules apply until age 65. The results suggest that policies aimed at extending the working lives of older individuals should not only address demand-side constraints but also facilitate transitions to new job arrangements later in life.

Keywords: Labor supply, Elderly employment, Pension reform, Retirement

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1 Introduction

Population ageing has become a pressing demographic and policy concern across advanced economies. Rising life expectancy, combined with persistently low fertility rates, averaging 2.3 globally and 1.6 across OECD countries (OECD, 2024), well below the replacement level of 2.1, has led to a rapid increase in the share of older populations. As a result, many economies face shrinking workforces and growing fiscal pressures. For example, the old-age dependency ratio now approaches one-third, meaning every three workers support one retiree, and public pension spending accounts for roughly 9 percent of GDP among OECD countries, exceeding 13 percent in some countries (OECD, 2021). These trends pose significant challenges for the sustainability of pension systems and public finances, making it increasingly important to extend the working lives of older individuals to boost tax revenues and contain pension expenditures.

To address ageing issues and encourage longer working lives, the pensionable age has recently been increased in many countries, such as the US (Mastrobuoni, 2009; Deshpande et al., 2024), the UK (Cribb et al., 2016; Blundell et al., 2016) and across Europe (Staubli and Zweimüller, 2013; Lalive et al., 2017; Geyer and Welteke, 2021; Rabaté et al., 2024). However, while this approach effectively delays retirement—most workers postpone retirement until they reach the new eligibility age—it is politically and practically challenging to keep raising the pensionable age. Moreover, little attention has been paid to how labor market regulations and firm practices around the pensionable age mediate retirement behaviors across different institutional contexts.

In this paper, we study how raising the pensionable age affects the labor supply of older workers in Japan, which is one of the world's most rapidly ageing societies and one with a highly strained pension system. Japan's old-age dependency ratio is approximately 70 percent, implying that fewer than two workers support one retiree. Using administrative tax records, we exploit a reform that gradually increased the eligibility age for earnings-related benefits (ERB), a key component of the public pension scheme, from 60 to 65 for males born after April 1st, 1951.[1] We focus on the first affected cohorts, whose ERB was delayed by one year

[1]Since more than half of females are not engaged in regular full-time employment in Japan, this paper focuses on males.

from age 60 to 61, and implement a dynamic difference-in-differences method that compares outcomes at the same ages for affected and unaffected birth cohorts to estimate the causal effect of the reform.

Japan offers a special institutional setting for studying the effects of pension age reforms. Unlike in many countries, where workers remain in their jobs while waiting for pension eligibility, Japanese workers are mandated to terminate employment contracts at age 60. Firms are legally obligated to offer re-employment to workers aged 60 to 65 under new contracts, often with different working conditions. Consequently, workers must adjust their labor supply at older ages within a framework that minimizes demand-side barriers. Given this setting, the reform's increase of pensionable age not only kept workers in the labor force, who would have otherwise stopped working and claimed ERB at age 60 before the reform, but also shifted workers into new employment contracts, allowing us to examine how firm practices and labor market regulations mediate workers' responses to the pension reform.

Our main finding is that Japanese workers remained persistently longer in the labor market following the one-year increase in the ERB eligibility age from 60 to 61. Specifically, the employment rate increased by approximately 5 percentage points among affected males, with this effect persisting until age 65. Although appearing modest in absolute magnitude, this effect is nevertheless large in Japanese context, where the employment rates were already high at late ages. Indeed, the reform induced roughly 60 percent of the workers who otherwise would have exited the labor force at age 60 to continue working. Moreover, not only did they continue working for the additional one year while the pension eligibility was delayed, but also for three additional years on average following this. To the best of our knowledge, such persistent employment effects—that is, working beyond the new pension eligibility age—have not been documented in previous studies, where individuals typically postpone retirement only until they become eligible again (Staubli and Zweimüller, 2013; Geyer and Welteke, 2021; Rabaté et al., 2024).

Importantly, Japanese workers are allowed to continue working after claiming pension benefits. We find that while affected cohorts persistently remained employed after age 60, they delayed claiming ERB only until age 61. Among individuals whose earnings at age 59 were in the first or second terciles,[2] ERB claims were delayed by approximately 0.9 year, consistent with compliance to the reform. Additionally, the increase in employment was homogeneous across individuals with different pre-60 earnings, suggesting that liquidity constraints were not the primary driver of extended employment in our settings. In fact, one year of post-60 earnings was sufficient to offset the temporary loss of pension income. Furthermore, despite the reform being announced as early as 2000, there is no evidence for anticipatory adjustments before age 60. Instead, our findings indicate that firm practices and institutional features jointly shaped individuals' responses to the reform, which we next proceed to document.

While the likelihood of continued employment was similar across income groups, heterogeneity emerged in job transitions at age 60. As noted above, in Japan, many workers need to terminate employment contracts typically at age 60 and resign new re-employment contracts to continue working. The re-employment contracts, normally provided by the firm that employed the worker at age 59, feature changes in working conditions, such as duties, working hours, and wages. We find that individuals with higher earnings at age 59 experienced sharper reductions in earnings after re-employment, suggesting variation in incentives to remain with previous employers. Lower earners tended to stay continuously employed with the same firm until age 65, while higher earners were more likely to switch firms. These patterns likely reflect both supply- and demand-side factors: from the supply side, lower earners might face greater difficulty finding new jobs from different firms, so they tend to accept the re-employment contracts from their age-59 firm, a choice which requires least effort on the workers' part, given the legally employment protection in place. From the demand side, higher earners were more likely to be employed by larger firms before age 60 and, subsequently, were rehired through other affiliated firms.

Since workers who switched firms after age 60 were more likely to adjust working conditions, we analyze the effects of the reform on their earnings after age 60 relative to their earnings at age 59. We find that lower earners were more likely to hold re-employment positions with earnings close to their prior levels, whereas higher earners experienced larger earnings declines. Overall, the heterogeneous response in job transitions indicates different incentives to continue employment across income groups. The results imply that policies to prolong employment of

[2]We excluded individuals whose earnings at age 59 were in the third tercile because their ages at first claim is not reliably measured due to the earnings test. More details will be explained in Section 3.3.

elderly workers beyond the pensionable age should both address demand-side constraints and encourage workers to explore new job arrangements at late ages.

This study contributes to several strands of literature. First, it adds to the literature examining the effects of increasing the pension eligibility age in Austria (Staubli and Zweimüller, 2013; Manoli and Weber, 2016) , in Germany (Geyer and Welteke, 2021; Etgeton et al., 2023), the US (Mastrobuoni, 2009; Deshpande et al., 2024), the UK (Cribb et al., 2016; Kanabar and Kalwij, 2022), Australia (Atalay and Barrett, 2015; Oguzoglu et al., 2020), Norway (Vestad, 2013; Johnsen et al., 2022), and Netherlands (Rabaté et al., 2024). We find large and persistent effects in the Japanese context, where the reform increased the labor supply of affected cohorts by approximately 0.29 years—comparable to the elasticity of 0.21 estimated in the Netherlands (Rabaté et al., 2024). While previous work has suggested that persistence is theoretically possible (Rabaté et al., 2024), empirical explanations have been limited. Our findings underscore the interplay between policy design, individual decision-making, and firm practices.

Second, we contribute to the literature on the determinants of retirement decisions. Previous studies have established the importance of financial incentives (Manoli and Weber, 2016; Blundell et al., 2016; Lalive et al., 2017; French et al., 2022; Lalive et al., 2023), framing effects (Behaghel and Blau, 2012; Seibold, 2021), and firm practices (Deshpande et al., 2024). Our results highlight how institutional features, especially the interaction of firm practices with labor market regulations, shape work–retirement decisions. Furthermore, our evidence aligns with emerging research on experimentation in labor economics. Previous studies show that experimenting with job search can lead to persistent changes in career paths (Belot et al., 2019; Bolhaar et al., 2019). Similarly, in the institutional setting of Japan, where workers were mandated to transition to new re-employment contracts at age 60 to continue working, the reform may prompt late-career experimentation with new job arrangements, fostering extended labor market participation beyond the pensionable age.

Lastly, this study also contributes to the literature on pension reform in Japan. Nakazawa (2025) focuses on the increase in the eligibility age of the flat-rate benefits in males from 2000 and finds that employment of elderly workers increases by 7–8 percent in cohorts affected by the reform. Kondo and Shigeoka (2017) examine the effects of the demand-side reforms that legally mandated employment protection. Nevertheless, to the best of our knowledge, we are the first to study the effects of raising the ERB eligibility age.

Moreover, compared to existing studies done in Japan, this study stands out by utilizing newly collected administrative data, allowing us to observe comprehensive income measures and accurate pension benefits information. Due to data availability, there were several points that remained ambiguous in existing studies. For instance, individuals may start receiving benefits as early as age 60 with some benefit reduction. Because data on pension claiming status has not been recorded in other Japanese social survey data, this channel has not been investigated in previous research. Using a new dataset that includes pension income, we found no evidence for this adjustment channel.

The rest of the paper is structured as follows. Section 2 describes the pension system in Japan. Section 3 shows the data and descriptive statistics. Section 4 outlines the identification strategy. Section 5 shows the effects of the reform on pension claiming and employment. Section 6 outlines the heterogeneity analysis in response to the reform by pre-60 earnings and Section 7 provides further discussion of potential explanations. Lastly, Section 8 concludes this study.

2 Background

In this section, we outline the institutional background relevant to our study. We begin with an overview of the public pension system in Japan. We then describe the reforms that increased the pension eligibility age, which serves as the focus of this study. Finally, we discuss other institutional settings, special in Japan, that shape work incentives at older ages.

2.1 Overview of the Public Pension System in Japan

Employee's Pension Insurance (EPI) is mandatory for all Japanese employees earning at least 88,000 JPY per month[3] and under age 70. Contributions equal 18.3 percent of earnings, shared evenly between employers and employees, and pension eligibility requires at least 25 years of contributions. The benefits received under this system are on average approximately

[3] Roughly, it includes those who work more than 20 hours per week.

61.7 percent of individuals' pre-retirement earnings (MHLW, 2019).

Specifically, the EPI benefits consist of two parts—the *flat-rate benefits* (FRB) and the *earnings-related benefits* (ERB)—with the latter being the focus of this study. The FRB depends solely on contribution years, while the ERB depends on both the average pre-retirement earnings and durations of participation. For a typical male with 40 years of contributions, the ERB accounts for roughly 60 percent of total pension benefits.[4].

Individuals may claim benefits as early as age 60 with an actuarial reduction. However, for cohorts eligible for ERB at age 61–64, delaying claiming ERB beyond the eligibility age would not increase the amount of ERB. Moreover, the additional pension premium from working during the ages of 60 to 64 would not be updated to the amount of ERB until age 65. Hence, there is little incentive to delay claiming for cohorts included in this study.

2.2 The Reform: Raising Pension Eligibility Age

Facing pressing aging issues, the Japanese government implemented a series of reforms to gradually raise the pension eligibility age in the early 2000s. These reforms began with a gradual increase in the FRB eligibility age in 2001, followed by a gradual increase in the ERB eligibility age in 2013 for males.[5]

The later set of reforms, which is the main focus of this study, raised the ERB eligibility age from 60 to 65 by one year every two years from 2013. The reform is still ongoing, and is expected to be completed in 2026. Figure 1 summarizes the ERB eligibility age by birth cohort, with the shaded area indicating the cohorts, born between Jan 1st, 1952, and March 31st, 1953, who were studied in this research. For all cohorts we examined in this study, they shared the same FRB eligibility age at 65. Hence, the ERB was the first pension benefit that influenced individuals' retirement decisions and this schedule provides quasi-experimental variation in pension eligibility between adjacent birth cohorts.

While individuals may also switch into relying on other social security programs, such as the Unemployment Insurance, or Disability Benefits (Staubli and Zweimüller, 2013; Geyer and Welteke, 2021) after the reform was introduced, this study focuses on the effects on

[4]More details about the Japanese pension system and the level of benefits are shown in Appendix B.1

[5]A similar reform was implemented for females, with the FRB eligibility age gradually increased from 2006, and the ERB eligibility age gradually increased from 2018. Since more than half of females were not regular (full-time) workers in Japan, we focus on males in this study.

employment since these benefits are not observed in our data.[6]

2.3 Labor Market Institutions Shaping Work Incentives

While this study focuses on the reform of increasing the ERB eligibility age, employment responses around the time of the pension eligibility age may also be influenced by firm practice and labor market regulations that existed before the reform is introduced in Japan. In this section, we provide a brief overview of these institutional details and how they might affect work incentives.

Firm Mandatory Retirement System Most Japanese workers may be involuntarily dismissed at the *firm mandatory retirement age*, an age unrelated to pension eligibility, when employers can legally terminate employment contracts for their employees. This system has been in place since 1994 for firms who hire employees under indefinite contracts.[7] Under this system, workers must re-sign contracts to continue working beyond the firm mandatory retirement age. As of 2021, about 80 percent of firms with 30 or more employees set the firm mandatory retirement age at 60. Consistent with this, Figure E1a shows the share of employed individuals under indefinite contracts remains between 80 and 90 percent until age 59, but declines sharply to about 40–50 percent after age 60.

Elderly Employment Stabilization Law While the firm mandatory retirement age is mostly set at 60, it is common for workers to be rehired by their former employers until age 65 due to the *Elderly Employment Stabilization Law* (EESL). Speficially, under this law, firms are required to implement one of the following three options: (i) abolish their firm mandatory retirement system, (ii) raise the firm mandatory retirement age to 65, or (iii) introduce a continuous employment system to rehire workers until age 65 (Kondo and Shigeoka, 2017).[8] In practice, most firms opted for the third option by rehiring workers through definite contracts (Figure E1b). As of 2023, 69.3 percent of firms rehired workers after the firm mandatory retirement age of 60 (MHLW, 2023b).

[6] More details regarding other social security schemes are presented in Appendix B.3.

[7] The indefinite contracts, also referred to "lifetime employment" contracts, were provided to regular workers, who were normally full-time workers, for most firms in Japan.

[8] The third option of EESL applies for workers born after April, 1946. Hence, the employment protection provided by EESL applies the same for all cohorts included in this study.

Moreover, despite the employment protection under EESL existing until age 65, the length of employment after age 60 is determined by workers. In practice, re-employment proceeds by renewing one-year definite contracts multiple times until age 65 (Figure E1c and Figure E1d). Therefore, workers have the flexibility to leave their jobs after they are rehired and before they turn age 65 if they so desire.

Earnings Test While the pension claiming behavior and employment status are independent in Japan, individuals will receive reduced pension benefits if they continue working after claiming due to the earning test.[9] While this creates a mild disincentive to work while claiming, the effect is minimal since delaying claiming does not increase the level of ERB for cohorts included in this analysis. However, it might cause measurement error for claiming age, a point we will discuss in Section 3.3.

Together, while the firm practice and labor market regulations applies to all the cohorts included in this analysis, these features create a unique environment for studying the effect of pension reforms (Table 1). Unlike other countries where individuals mainly remain at their existing job while waiting for the delayed pension, such as UK (Cribb et al., 2016) and US (Mastrobuoni, 2009), Japanese workers need to change employment contracts to stay employed at age 60, the former pension eligibility age. The reform not only changed the financial incentives to work, but also forced individuals to proceed with new contracts, which they would have not chosen if they had pension income at earlier ages. We will further discuss this point in Section 5.4.

3 Data and Descriptive Statistics

This section presents the data set used, outlines the sample construction process, variables of interest, and summarizes the descriptive statistics.

[9]The pension benefits are reduced when the combined monthly earnings and original pension before the earnings test exceed a certain threshold. More details are shown in Appendix B.4.

3.1 Data

We employed the linked residential tax records and registry data from five municipalities in Japan as our main data set. The data is collected by the Center for Research and Education in Program Evaluation (CREPE) at the University of Tokyo. The tax records include variables related to earnings and pension benefits, making it possible to examine labor supply adjustment in comprehensive ways. These records are linked with registry data that contains demographic characteristics such as sex, year and month of birth, and relationship to the household head. The registry data includes all individuals residing in the municipality who were alive on January 1st of each year, regardless of their working status.

This dataset allows us to perform novel analysis in several dimensions. First, relative to other survey data, it covers more recent periods, which is essential to study the on-going pension reform. Second, it provides longer coverage of residents in Japan, enabling us to examine responses over relatively long run. Third, whereas existing government statistics are primarily cross-sectional, it constitutes the first panel dataset from administrative records. Lastly, it contains detailed measures of income and pension benefits, allowing for a comprehensive examination of labor supply adjustments.

3.2 Sample Construction

We considered the sample of males born between January 1st, 1952 and March 31st, 1955. To minimize the effects of the FRB reform, we restricted the sample to individuals born after April, 1951 as they share a uniform FRB eligibility age of 65, but maintain some variation in the ERB eligibility ages. Moreover, to focus on cohorts who were continuously observed from age 59 to 66, we also excluded individuals born before January 1st, 1952, as they were not observed at age 59. In addition, we excluded individuals born after March 31th, 1955, as they turned age 66 later than 2021, thereby minimizing exposure to the effects of the pandemic. to proxy for those entitled to ERB and hence likely to be affected by the reform

Moreover, to proxy for those entitled to ERB and hence likely to be affected by the reform, we further restricted the sample to those who had annual earnings at or above 1.056 million JPY (88,000 JPY $\times 12$) at age 59 (See Section 2.1). This means we only focused on workers who were still active in the labor market one year prior to the pre-reform ERB eligibility age. One limitation of this approach is that we may have inadvertently excluded individuals who were eligible for ERB but exited the labor force at age 59 or experienced employment gaps within that year, leading to reduced annual income. As a robustness check, we re-estimated our results including individuals whose earnings are below the threshold and found no substantial differences (See Section 5.3.1). Lastly, we excluded individuals in the top 1 percent of earnings to minimize the influence of outliers. The final sample consists of 1,593 unique male individuals from five municipalities, observed over eleven years (Table D1).

One may be concerned that five municipalities are not representative of the Japanese population. However, our results suggest this is unlikely to be the case. In terms of observable characteristics, these municipalities are broadly similar to the national average in industry composition, and slightly below average in the share of elderly population (Figure E2). In addition, we conducted a leave-one-out exercise, re-estimating the effects on employment while excluding each municipality one at a time to show that the employment effect is not driven by a specific municipality. The resulting effect patterns were consistent with our main findings, with effect sizes remaining comparable (Figure E3).

3.3 Variable Construction

This subsection explains how variables of interest were constructed, and outlines some limitations of each variable.

Pension Claim Our measurement of pension claiming is based on whether individuals report positive pension benefits in a tax year. The first claiming age refers to the age at which an individual initially begins to receive pension benefits, measured in the calendar year they attain that age.

Two measurement errors may occur for the claiming age. First, the public pension benefits observed in the tax data include other pension benefits, such as disability pension and firm pension. Since the share of the population eligible for disability pension is small, and the firm pension is provided by employers and cannot be opted out by employees,[10] the effects from other benefits should be minimal. Second, the pension amount reported in the tax record is

[10]The number of individuals receiving disability pension account for only approximately 2 percent of the population in 2025. More institutional details are listed in Appendix B.4.

the net amount received after the earnings test, this measure may under-report the actual pension claim status if their pension benefits were exhausted. This will mainly affect high-earning individuals, and hence in our main analysis we will exclude people with earnings in the third-tercile.

Employment Since employment status and the exact month that individual quit jobs is not observed in the tax data, individuals are classified as employed if they reported positive earnings, measured in the following calender year when they attain that age.[11]

3.4 Descriptive Statistics

In this section, we document overall employment and claiming patterns across ages in the data. To minimize compositional effects, we restrict this analysis to the four municipalities where information is available for the oldest unaffected cohorts at age 59.

3.4.1 Pension Claim

Most individuals claim ERB when they become eligible, with some delaying claiming to the following year (Figure 2a).[12] Specifically, approximately 60 percent of individuals claim their benefits at the age they become eligible, while about 20 percent of them claim in the following year.

One year after reaching the ERB eligibility age, approximately 80 percent of individuals in both the unaffected and affected cohorts claim their pension (Figure 2a). Note that not all individuals appear to claim. This is because we did not observe ERB eligibility directly, so some residents, who were not eligible for ERB but had total monthly earnings at or above the threshold for EPI (88,000 JPY), were included in the sample (See Section 3.2).

Moreover, the share of individuals claiming pension benefits remains below one even after reaching age 66, when all residents in Japan were eligible for FRB (Figure 2a). This could

[11]For example, consider an individual who quits working and claims their pension in the month of their 60th birthday. In the calender year after he turns 60, we observe positive earnings, which reflects income from earlier months prior to his 60th birthday, and positive pension benefits, which reflects the pension income from later months after he turns 60. Hence, the actual employment status is reflected in the following calender year when he turns 60.

[12]This step-wise claiming pattern is because some Japanese workers would aligned their pension claiming with the fiscal year, running from April 1st to the March 31st in the following calender year, especially for those born in the forth quarter. This suggests seasonality in claiming behaviors. More details are listed in Appendix B.2.

be because some individuals who were only eligible for FRB postponed claiming, or some individuals fully exhausted their pension benefits due to the earnings test.

3.4.2 Employment

The employment rates decline gradually after reaching age 60, with the unaffected cohorts decreasing more sharply at age 60 compared to the affected cohorts (Figure 2b). Specifically, the employment rate for unaffected cohorts declines approximately 8 percentage points at age 60, while this number is smaller at approximately 4 percentage points for affected cohorts (Figure 2b).

After reaching age 61, the employment rates continue to decline gradually for both cohorts, with the difference in employment rates between the two cohorts persisting until age 66. Moreover, the employment rate remains at about 70 to 75 percent for both cohorts after reaching age 66, indicating high labor market attachment for EPI insurers (Figure 2b).

4 Methodology

Our identification strategy relies on the comparison between cohorts with different ERB eligibility ages. For time-invariant dependent variables such as the claiming age, we employed the following specification:

$$Y_i = \alpha + \beta D_{c(i)} + \eta_{p(i)} + MOB_i + \epsilon_i, \tag{1}$$

where Y_i represents the time-invariant outcomes for individual i. $D_{c(i)}$ is an indicator variable indicating whether individuals born in cohort c are affected by the reform. $\eta_{p(i)}$ and MOB_i are municipality fixed effects and month of birth fixed effects, respectively. To account for the correlation among observations for individuals born in the same year and quarter of birth, the standard errors are clustered at the year-month birth cohort level.

Moreover, for outcomes that evolve with age, we estimate a specification that compares outcomes across cohorts at each age using the following equation:

$$Y_{ia} = \alpha + \sum_{j,j\neq 59}^{66} \beta^j D_{c(i)} \cdot \mathbf{1}\{a = j\} + \mathbf{X}_{ia}\theta + \gamma_a + \lambda_c + \epsilon_{ia}, \tag{2}$$

where Y_{ia} represents the outcome for individual i born in cohort $c(i)$ at age a. γ_a and λ_c denote age fixed effects and cohort effects. $\mathbf{X}_{ia}$ includes a set of control variables, such as the municipality fixed effects, month of birth fixed effects, and municipality and month of birth-specific age fixed effects, as well as the ratio of job to applications observed at age a in the municipality of individual i. Standard errors are clustered at the year-month birth cohort level. The reference age is set at age 59. Hence, β^j estimates the effect of reform on the outcome at age j relative to age 59.

The identifying assumption is that Y_{ia} would be comparable between the affected cohorts and unaffected cohorts at the same age if the reform had not been implemented, after controlling for cohort fixed effects and other controls.

One potential concern is that the estimated employment effect may stem from cohort-specific characteristics unrelated to the ERB eligibility. To investigate this potential threat, we first show the raw data trends in employment across ages for all cohorts with the same FRB eligibility age. Then, we formally test for differences across cohorts with the same ERB and FRB eligibility ages.

The employment trends are similar across fiscal-year-of-birth cohorts who had the same ERB and FRB eligibility ages (Figure 3a). Specifically, the employment rate is approximately 80 percent before age 60 for all cohorts in regardless of ERB eligibility age (Figure 3a). Additionally, for affected cohorts born in fiscal year 1953 to 1955, the decline in employment after age 60 is slower than unaffected cohorts born in fiscal year 1951 and 1952.[13]

To formally examine the possibility of cohort-specific effects, we replaced $D_{c(i)}$ in Equation (2) with the birth-cohort specific dummies and re-estimated the equation using individuals who shared the same ERB and FRB eligibility ages. This specification can be interpreted as placebo tests, as it estimates the effect of the reform hypothetically applied to cohorts that was not affected. Figure 3b shows the results of this estimation: there is no significant difference in employment trajectories between birth cohorts with the same ERB and FRB eligibility ages.

[13]Note that we do not restrict the sample using earnings at age 59 in this exercise as it is unobservable for individuals born in 1951.

5 Effects of Pension Reform on Workers

In this section, we show the impact of increasing the ERB eligibility age from 60 to 61 on claiming behaviors and employment. Specifically, we start by reporting the baseline effects on claiming age and employment to confirm the compliance to the reform using the main sample. Then, we examine the effects using alternative samples, specifications and measure of labor supply for robustness checks.

5.1 Effects on Claiming Age

First, we examine the effects of increasing the ERB eligibility age on when individuals first claim their pension (Table 2). Facing a one year increase in the ERB eligibility age, affected males delayed pension claiming by approximately 0.9 years (Column 1 in Table 2 Panel A), indicating compliance with the pension reform.[14]

Additionally, there is no significant difference in the effect size between affected males who earned a first-tercile and those who earned a second-tercile income at age 59, indicating no evidence of liquidity constraints (Column 2 and 3 in Table 2 Panel A).

Since the employment status is independent of pension claiming, postponing claiming pension does not necessarily imply an increase in employment under our context. In the next section, we examine the effect of the higher ERB eligibility age on employment.

5.2 Effects on Employment

Compared to the cohorts unaffected by the reform, individuals whose ERB eligibility age was increased to 61 years exhibited approximately a 4 to 6 percentage points higher employment rate, with the effect persistent even beyond the new ERB eligibility age (Figure 4a).

The persistent employment effects beyond the new ERB eligibility age observed in this study is uncommon in the literature. Most studies find employment effects disappear once individuals become eligible for pension benefits (Mastrobuoni, 2009; Staubli and Zweimüller,

[14]We excluded individuals who earned a third-tercile income at age 59 in this analysis for whom the age when individuals first claimed pension may be not reliable measured. Specifically, individuals with higher earnings might exhaust their ERB due to the earnings test, and start to report positive pension after their earnings reduced as they aged. This might underestimate the effects since they were more likely to be able to afford delaying ERB longer.

2013; Cribb et al., 2016; Geyer and Welteke, 2021; Rabaté et al., 2024), with some findings indicating that individuals even leave their jobs at the original eligibility age (Deshpande et al., 2024). On the contrary, we found that individuals in Japan remain employed even after they become eligible for the ERB again. While Rabaté et al. (2024) discuss the possibility for persistent employment effects, referring to "downstream effects", there is little explanation for why this might occur. In Section 6, we will discuss how firm practices and labor market regulations jointly shape the employment responses observed in this study.

Moreover, while the younger affected cohorts who faced longer delay in pension, we found homogeneous employment effects in terms of persistent effect pattern and effect size for them (Figure 4b). Atalay and Barrett (2015) found that younger cohorts, who faced longer delayed in pension eligibility, may experience larger changes in pension wealth, hence they may respond more strongly to the reform of increasing the eligibility age in Australia.[15] However, our results suggested similar response to the reform among cohorts faced different length of delay in pension benefits, indicating that the observed effect may primarily driven by the changes in labor supply occurred at the age of 60, when individuals started to be influenced by the firm practices and legal employment protection.

Furthermore, while the short-term effect on employment we observe in absolute terms is comparable to other estimates in the literature (Cribb et al., 2016; Staubli and Zweimüller, 2013),[16] this drastically understates the true size of the effect. Specifically, before the reform, only 8 percentage points of individuals in the unaffected cohorts quit labor force at age 60. Therefore, a roughly 5-percentage-point employment effect indicates that the reform resulted in approximately 60 percent of individuals who would have retired remaining in employment. Additionally, the reform effectively extended the labor supply over the whole affected cohorts by 0.29 years.[17] While this number is comparable to the elasticity of 0.21 found for a reform increasing the statutory retirement age in the Netherlands (Rabaté et al., 2024), it dramatically understates the real magnitude of the effect we observed in this study. Among those who would

[15] On the other hand, since younger affected cohorts, who faced longer delay in pension, might respond less strongly from the older affected cohorts because they had longer time to adjust (Staubli and Zweimüller, 2013; Nakazawa, 2025). Since the reform was announced more than a decade before its implementation, the effect through this channel should be minimal.

[16] For example, following an increase in pension eligibility age, employment increased by 6.3 percentage points among females in UK (Cribb et al., 2016), by 9.8 percentage points for Austrian males (Staubli and Zweimüller, 2013), and by 20–22 percentage points for Dutch men (Rabaté et al., 2024).

[17] The calculation of effective extended labor supply is detailed in Appendix A.

have become non-employment, roughly 60 percent of them stayed longer in the labor force and they remained in employment for approximately 3 years longer on average.

Lastly, we found no evidence for the anticipation effect, although the policy was announced more than a decade before its implementation. Individuals might change their labor supply even before they reach the former eligibility age, so called upstream effect (Rabaté et al., 2024); however, we found no significant evidence that affected individuals had a higher employment rate than the unaffected ones before age 60. Consider the analysis for our main sample in Figure 4a. At age 58, the affected cohorts appeared not to have a higher employment rate at a statistically significant level. This result is similar to other empirical findings where the pension reform is announced a long time before implementation (Mastrobuoni, 2009; Geyer and Welteke, 2021; Deshpande et al., 2024).

5.3 Robustness Checks

In this subsection, we outline the robustness checks for our results using an alternative sample, specification, and measure of labor supply. Specifically, we first show results including those who earned below 88,000 JPY per month at age 59. Second, we estimate the effects using a Regression Discontinuity Design (RDD). Lastly, we estimate the effects of the reform on the probability of being continuously employed.

5.3.1 Alternative Sample

Recall that not all individuals are eligible for ERB, and those ineligible would not be affected by the reform. Since we do not directly observe ERB eligibility, we proxied for it in our main analysis by restricting to individuals who earned at or above 88,000 JPY at age 59. However, individuals who either stopped working prior to age 59 or did not work for some months of that year[18] might have been eligible for ERB, but misclassified as ineligible. Excluding these individuals may lead to ambiguous bias for our results.

As a robustness check, we included all individuals in regardless of their level of earnings and re-estimated our main results. For our results on claiming age, the effect size after including

[18]Specifically, Individuals took some months off in the year turning 59 years old and had the annual earnings lower than 1.056 million JPY (88,000 $\times$ 12), despite being eligible for ERB, would be excluded from our main sample.

those who earned below 88,000 JPY was consistent with the main analysis (Column 1 in Table 2 Panel B). The effect size was insignificantly smaller for the first-tercile group than baseline results. However, this is because the average claiming age for the unaffected cohorts was higher after including those who were only eligible for FRB at age 65 (Column 2 in Table 2 Panel B).

The employment results for this larger sample were also broadly consistent with the main analysis, as shown in Figure E4a. Moreover, our results showed that the effect on employment was consistent in size and persistence with the baseline results even after including the individuals who may have been incorrectly excluded as ineligible for ERB.

5.3.2 Alternative Specification

In the main specification, we estimated the effects of increasing the ERB eligibility age by comparing the employment rate between affected and unaffected cohorts. However, individuals born just after the cutoff date, April 1st, 1953, may be more directly affected by the reform than those born further away from the threshold. To account for this possibility, we varied the subsamples with bandwidth around the cutoff to 9 and 12 months, with the bandwidth of 24 months equivalent to sample in the baseline specification, and estimated the following equation:

$$Y_i^a = \alpha + \beta^a \mathbf{1}\{\text{birth date}_i > \text{cutoff date}\} + \eta_{p(i)} + MOB_i + \epsilon_i, \tag{3}$$

where Y_i^a is the age-specific employment for individual i. The rest notation is the same as equation (1). β^a captures the effect of increasing the ERB eligibility age on employment at age a. We estimated Equation (3) separately for the ages of 58 to 65 and plotted the coefficients across ages in Figure E4b.

We found no evidence that the effects were larger for those born closer to the cutoff date. The estimates using different bandwidths suggested similar employment effects of about 5 percentage points (Figure E4b). The homogeneous responses for cohorts could be because the reform was announced more than a decade earlier than it was implemented. As a result, affected males with the same ERB eligibility age would have enough time to adjust their responses regardless of the distance between their birth date and the cutoff date.

5.3.3 Alternative Measure of Labor Supply

In the main analysis, we have used as a measure of employment as an indicator variable defined by whether the individual reported positive earnings in a particular calender year. However, this measure may not necessarily reflect continuous employment of the same individual over time because it cannot distinguish between continuous employment and individuals moving in and out of employment across years. To address this, we examine the effect of the reform on the probability of being continuously employed from age 59 onwards using equation (1). Using this alternative measure, we found that the results observed in Figure 4a were indeed driven by continuous employment, as individuals in the affected cohorts were more likely to remain employed from age 59 onwards (Table D2).

5.4 Changes in Working Conditions after Age 60

In the previous section, we have shown the effects of the reform on the probability of having positive earnings in a particular year. However, this measure may not capture the changes in labor supply at the intensive margin.

Importantly, while employers are required to rehire workers under the EESL, they are not required to offer the same contract as was in place before the firm mandatory retirement age. Among firms that introduced the continuous employment system, 64 percent of them only re-employed workers under new fixed-term contracts, normally under altered conditions, such as changed hourly wage, working hours, and job responsibilities (Kajitani, 2006; OECD, 2018).

Our evidence suggested that earnings reduced substantially after age 60, as shown in Figure E7a.[19] Specifically, conditional on employment, annual earnings reduced by approximately 20 to 45 percent during the ages of 60 to 61.[20] Additionally, after age 61, there is no further reduction in earnings, indicating that the same (or similar) contract was subsequently renewed.

Hence, the reform not only kept workers in employment, who would have retired at age 60 if the reform was not introduced, but also shifted individuals into jobs with different working

[19] One may be concerned that individuals in the affected cohorts, while still employed, might reduce their earnings before they turned age 60. Examining the change in annual earnings for the younger affected cohorts, we found no evidence that annual earnings reduced before age 60, indicating no upstream effects in employment at the intensive margin (Figure E6).

[20] The reduction in annual earnings could be driven by change in either hourly wage or working hours. Evidence suggested that the hourly wage reduced by approximately 20 to 30 percent after age 60, as shown in Figure E7b.

conditions.

6 Heterogeneity Analysis by Earnings at Age 59

We have shown that affected individuals remained in the labor force even after reaching the new eligibility age. In this section, we examine whether these effects on labor market outcomes differ by earnings at age 59. We start by examining the effects on the probability of having a job. Second, we focus on the effects on job transitions after age 60. Lastly, we discuss on the effects on the probability of having jobs with different earnings compared to age 59.

A common expectation is that individuals with lower income would respond more strongly to the reform due to liquidity constraints. However, we find no evidence that employment responses differed by earnings at age 59, as shown in Figure 5a. This result is similar to Geyer and Welteke (2021), who found no heterogeneity across the income distribution in women's employment after increasing the early retirement age in Germany.

However, evidence suggested that the reduction in earnings differs across income groups. Conditional on employment, individuals with higher earnings at age 59 experienced a earnings decline of approximately 40 to 60 percent between the ages of 60 to 61 (Figure E7d). In contrast, those with lower earnings at age 59 experienced a smaller reduction of about 20 to 40 percent (Figure E7c). Because these changes in earnings might be primarily driven by contract adjustments at the firm mandatory retirement age, the observed pattern of post-60 earnings loss may suggest different incentives to remain employed with the same firm at age 59 across income groups.

To address these issues, we examine the effects of the reform on the probability of being employed by the same firm at age 59 for different income groups. Evidence suggested that among affected cohorts, individuals with lower earnings at age 59 were more likely to remain employed with their former employer (Figure 5b), indicating that lower earners were more inclined to utilize the employment protection provided by their employer at age 59 after the reform.

However, we find no similar responses for among higher earners (Figure 5b). That is, the reform did not seem to affect whether higher earners remain with their previous employer or not, only whether they remain employed at all and at what earnings. This difference could be stemming from both the labor supply side and demand side. On the supply side, higher earners might find it easier to obtain employment with other firms after age 60. On the demand side, they might be employed by large Japanese firms at age 59 and subsequently rehired indirectly through affiliated firms.[21]

Moreover, since workers were not necessarily rehired with the same working conditions as they had prior to age 60, we examine the effects of the reform on the probability of being employed with different relative earnings to age 59 (Figure 5c). We found that lower earners in the affected cohorts were more likely remain employed with more than 70 percent of their earnings at age 59, while higher earners in the affected cohorts had no similar response. These results still hold if we increase the threshold of earnings relative to age 59 (Figure E10).

7 Discussions of Potential Explanations

In this section, we discuss potential explanations behind the persistent increase in employment as a result of the reform. We argue that compensating for foregone pension income is unlikely to be the main driver; instead, behavioral channels provide a more plausible explanation.

Recovering Monetary Loss Since the pension reform is not actuarially fair – that is, the higher eligibility age effectively translated into a permanent loss in lifetime income – one possible explanation for continued working beyond the post-reform ERB eligibility age is that individuals seek to offset the foregone pension income. Specifically, the loss of one year of pension benefits may have been sufficiently large that workers needed several additional years of work to restore their lifetime income to the pre-reform trajectory.

However, a simple comparison of ERB and earnings suggests this mechanism is unlikely. For workers between the 25th and 75th percentiles of the income distribution at age 59, the average annual pension before the earnings test was 924,564 JPY, while the average post-60 salary was 2,944,466 JPY.[22] This implies that even in the extreme case of having no savings before age 60, the loss of one year of pension benefits could be recovered in less than 0.314 years

[21] In Japan, larger firms often rehire workers after the firm mandatory retirement age through affiliated firms or subsidiaries, a practice known as *shukko* or *tenseki* (MHLW, 2023a). Approximately half of large firms adopt these arrangements to flexibly adjust workers' roles and wages (JILPT, 2018).

[22] The earnings were computed at age 61 for affected cohorts.

of additional work, without considering the future increase in pension wealth after age 65.[23] Therefore, it is unlikely that individuals remained in the labor market solely to compensate for the pension income loss.

Peer Effects Previous studies have suggested that individuals make retirement decisions based on factors apart from monetary incentives, and a range of behavioral effects on retirement behaviors (Chan and Stevens, 2008; Liebman and Luttmer, 2015; Hentall-MacCuish, 2024). Workplace practices may play an important role. For example, Deshpande et al. (2024) finds that in the United States, although the pension claiming age is 66, many individuals retire at 65. The study attributes this result to workplace practice, specifically, employees working at firms with a higher share of colleagues retiring at 65 are themselves more likely to retire at 65 rather than at 66.

A similar mechanism may operate in our setting. Individuals could be more inclined to continue working if they observe many of their peers remaining employed beyond age 60. This would imply stronger employment effects in firms with a higher share of older employees. While our data does not cover the full employee composition within firms, we proxy for peers with stronger labor market attachment using share of employees aged above 60 within the same establishment.[24] Consistent with this hypothesis, we find that individuals in these firms are more likely to remain employed for longer (Figure 6).

Status Quo Bias Under the EESL, firms are legally required to offer re-employment contracts to workers who wished to continue until age 65 (See Section 2.3). Although workers retained the option to retire earlier than age 65 once re-employed, they may prefer to remain employed until the final year covered by the EESL, consistent with status quo bias (Samuelson and Zeckhauser, 1988; Kahneman et al., 1991). This interpretation is plausible in our context, especially for lower earners who were inclined to remain employed with their former firm at age 59 (Figure 5b).

[23]To be precise, the lost pension benefits could be recovered even faster than 924,564/2,944,466 years, since continued work also generates additional pension contributions, leading to a higher pension wealth. Figure E12 shows the distribution of earnings at age 61 and the pension amount before the earnings test for affected males who were employed and claimed pension at age 61.

[24]This proxy measure assumes that the choice of whether to reside in the same municipality of the firm (and hence be observed in our dataset) is independent of age.

Experimentation in Working Experience Unlike other countries where workers often stayed in their existing jobs while waiting for delayed pension, Japanese workers normally need to resign employment contracts if they continue working after age 60. Additionally, Japanese firms typically offered re-employment with lighter duties, shorter hours, and lower wages, especially in larger firms (Kajitani, 2006; OECD, 2018). Hence, it is plausible that, in the Japanese context, workers in affected cohorts experienced new job conditions while awaiting their pension, which leads to them learning about employment possibilities that they would not otherwise have considered. It might have also affected their choice of whether to continue employment after pension eligibility, and hence reshaped their long-run labor supply.[25] This is consistent with what we observed, in particular that higher earners who experienced larger changes in their working conditions (in terms of firm and earnings) tended to stay longer in the labor market (Figure 5b).

8 Conclusion

This paper studies how raising the pensionable age in Japan affects the labor supply of older workers in Japan. Leveraging Japan's unique institutional setting, where workers' contracts are terminated at age 60 and firms are forced to offer reemployment opportunities from age 60 to 65, we analyze how firm practices and labor market regulations jointly shaped responses to the pension reform.

The main finding is that Japanese workers whose earnings-related-benefits (ERB) were postponed by one year at age 60 typically maintained employment persistently until age 66, which is a novel finding in this literature. Specifically, 60 percent of individuals who would have gone into non-employment stayed longer in the labor force, indicating an increase in length of their labor supply of 3 years.

Because workers are allowed to continue working after age 60, we also find that individuals delayed claiming ERB by approximately 0.9 years, suggesting strong compliance with the

[25] A growing literature highlights the role of experimentation in experience for explanation individuals' hysteresis responses: temporary exposure to new conditions can lead to persistent behavioral change (Larcom et al., 2017; Bolhaar et al., 2019; Saez et al., 2021). For example, Larcom et al. (2017) show that many London commuters permanently adopted new travel routes discovered during a temporary tube strike. Similar evidence has been documented for job-search behavior (Bolhaar et al., 2019) and hiring subsidies (Saez et al., 2021) following temporary changes in labor market policies.

pension reform. Moreover, despite the reform being announced more than a decade ago prior to implementation, there is no evidence of anticipatory responses.

Furthermore, while the increase in employment rates was homogeneous among different income groups, which suggested that liquidity constraints were not the main driver for persistent employment, responses at the transitions age of 60 differed. Lower earners extended employment primarily by accepting re-employment contracts from their former employers, while higher earners—facing larger earnings reduction—might have different jobs after age 60.

Additionally, workers whose firm at age 59 had a higher share of employees aged 60 and above were more likely to persistently stay longer in the labor force. This may reflect that these firms provided more suitable jobs for older workers, or that peer effects encouraged continued employment beyond age 60.

Overall, these findings highlight how firm practices and labor market regulations jointly shape responses to the pension reform. To prolong the employment of elder workers beyond the pensionable age, policymakers should both address demand-side constraints and encourage workers to explore new job arrangements at late ages.

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Tables

Table 1: Employment and Pension Options across Ages

Age	Before reform: ERB eligible at 60	After reform: ERB eligible at 61
59	employment under indefinite contract.	Same as before
60	Mandatory retirement • Option 1: quit working, claim ERB • Option 2: re-employed, claim ERB, reduced amount after the earnings test	Mandatory retirement • Option 1: quit working without pension • Option 2: re-employed without pension
61	Same option as age 60	**Newly eligible** for ERB. • Option 1: quit working, claim ERB • Option 2: re-employed, claim ERB, reduced amount after the earnings test
62–64	Same as before	Same as before
65	Re-employment not available any more	Re-employment not available any more

Notes: ERB = Earnings-Related Benefit component of Japan's Employees' Pension Insurance (EPI). The firm mandatory retirement age is set at 60 for most firms; re-employment contracts are renewable until 65. The 2013 reform increased ERB eligibility by one year for men born April 1953–March 1955.

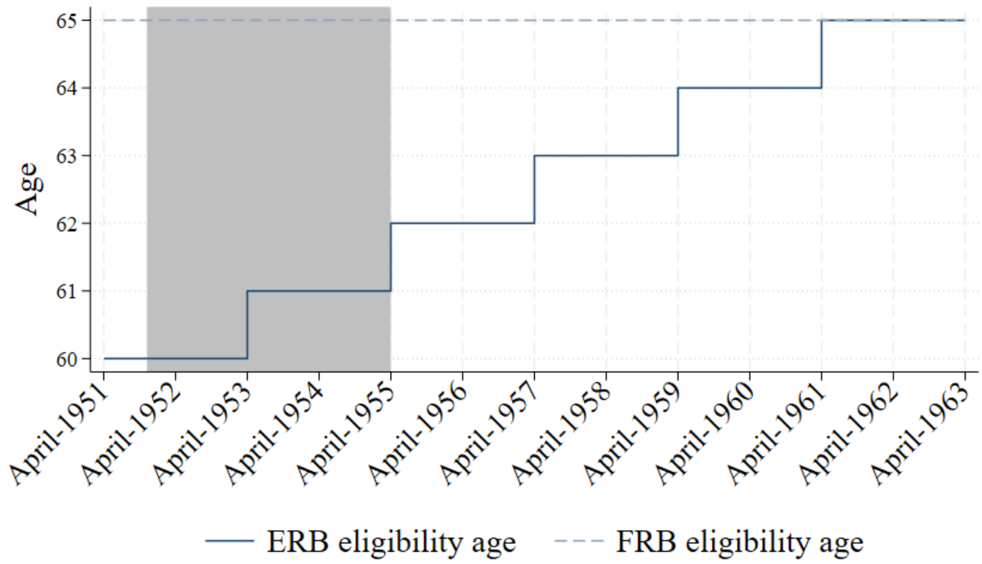
Table 2: Effects of increasing ERB eligibility age from 60 to 61 on first claiming age

	(1) Claim age	(2) Claim age	(3) Claim age
Panel A: Baseline results			
1[ERB eligible at 61]	0.934***	0.859***	0.996***
	(0.113)	(0.125)	(0.167)
Mean claiming age: control	60.618	60.486	60.729
Observations	1,035	519	516
Panel B: Including those with monthly earnings < 88,000JPY			
1[ERB eligible at 61]	0.805***	0.677***	0.986***
	(0.097)	(0.138)	(0.110)
Mean claiming age: control	60.731	60.982	60.467
Observations	1,399	700	699
Subsample by earnings at age 59	All	Low	Median
Month of birth FE	YES	YES	YES
Municipality FE	YES	YES	YES

Notes: The table shows the effects of increasing ERB eligibility age from 60 to 61 on first claiming age for males. The first claiming age is defined as the calendar year in which individuals first report positive pension benefits. Panel A shows the baseline results using the main sample; Panel B shows the robustness check including those who had monthly earnings below 88,000 JPY. Column 1 shows the aggregated effects for the main sample. Columns 2 to 4 show the effects for subsamples divided by earnings at age 59 conditional on year and municipality fixed effects. The main sample is restricted to only male individuals, who were observed in the calender year in which they turned age 59–66, and had monthly earnings at least 88,000 JPY at age 59. We further restricted the sample to those who earned a first or second tercile income at age 59 so that our measure of first claiming age was more reliable. Low group indicates for those who earned a first tercile income at age 59, and median group indicates for those who earned a second tercile at age 59. Robust Standard Errors are clustered at the year-month birth cohort level. Significance levels are $^{*}p < 0.1$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

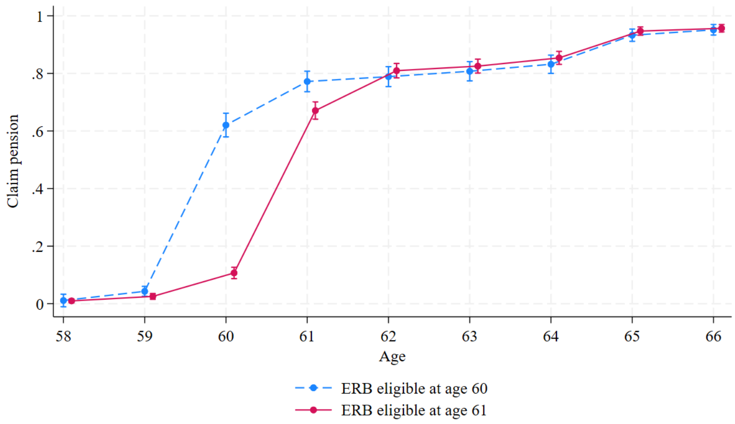
Figures

Figure 1: Pension eligibility ages by birth cohorts

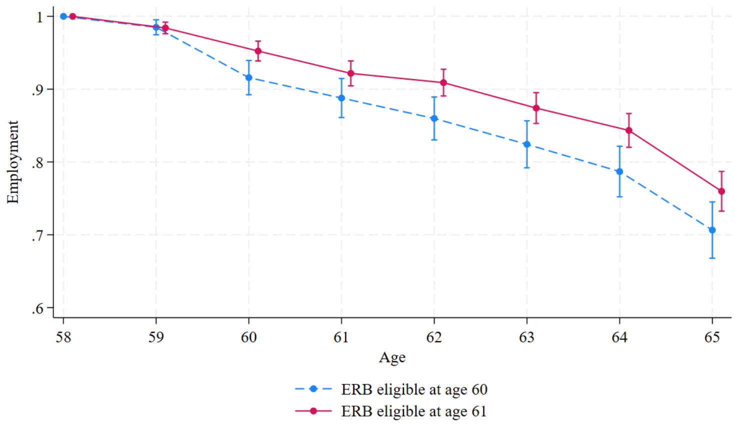


Notes: This figure shows the pension eligibility age by birth cohorts for males. The shaded area includes cohorts used in the main sample. This figure is generated by the authors using government statistics.

Figure 2: Pension claim rates and employment by age



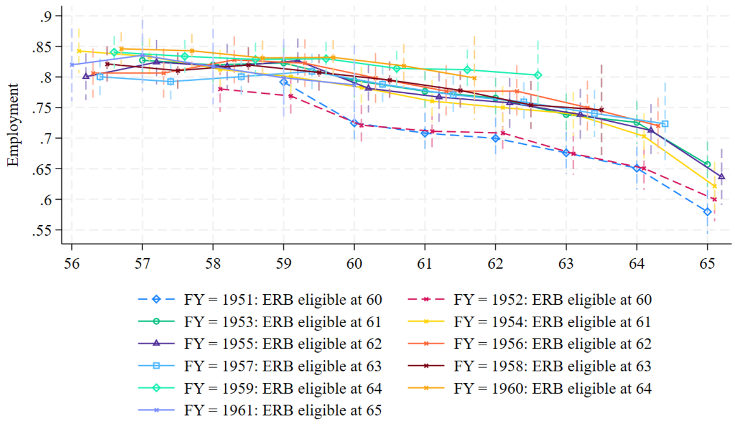
(a) Pension claim



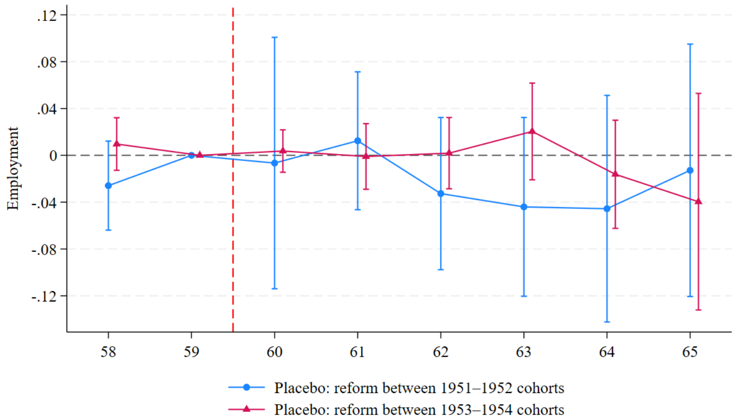
(b) Employment

Notes: Figure a (b) displays the claiming rate (employment) by age and cohorts for males. Claiming pension is measured as individuals who report positive pension benefits in the calender year they turn a given age. Employment is measured by whether individuals earn positive salary income in the following calender year they began at the stated age. The sample is restricted to only male individuals, who were observed in the calender year they turned age 59–66, had monthly earnings at least 88,000 JPY at age 59, and resided in municipalities where the oldest unaffected cohorts have available information at age 59. Confidence intervals shown denote 95% confidence level.

Figure 3: Checking for cohort-specific trends



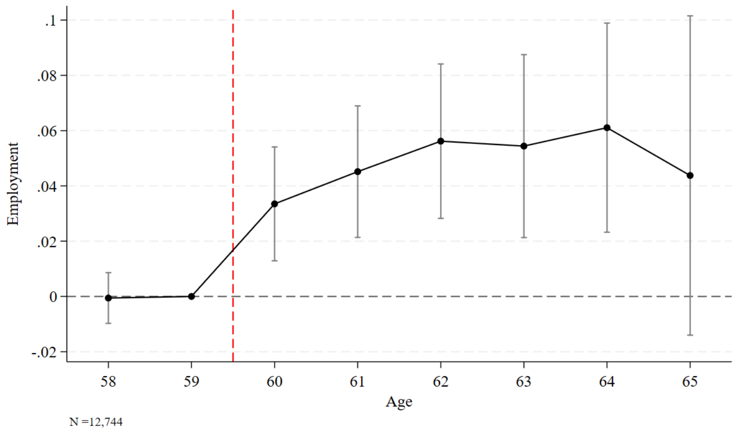
(a) Employment rate by age: by fiscal year of birth



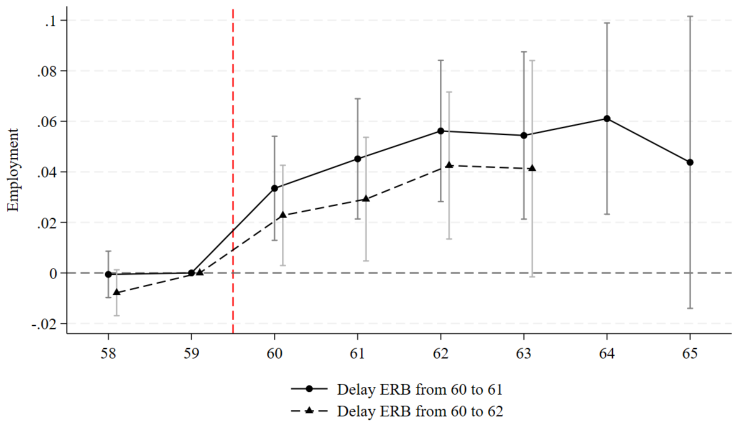
(b) Placebo treatment

Notes: Figure a shows the raw data trend in employment over age, split by birth cohort (defined by fiscal year as described in the main text) for males. Figure b shows the the result of a placebo treatment for males with the same ERB eligibility age. The placebo treatment is comparing the employment at the same age between two birth cohorts who shared the same earnings-related-benefits eligibility age (ERB). Employment is measured by whether individuals earn positive salary income in the following calender year they began at the stated age. The sample is restricted to only male individuals born after April 1st, 1951, who were continuously observed in the calender year they turned the stated age. Confidence intervals shown denote 95% confidence level.

Figure 4: Employment effects of raising ERB eligibility age



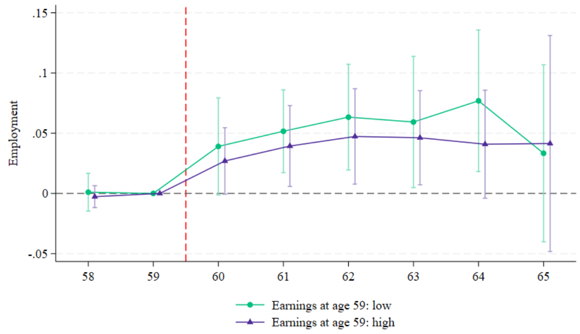
(a) From age 60 to 61



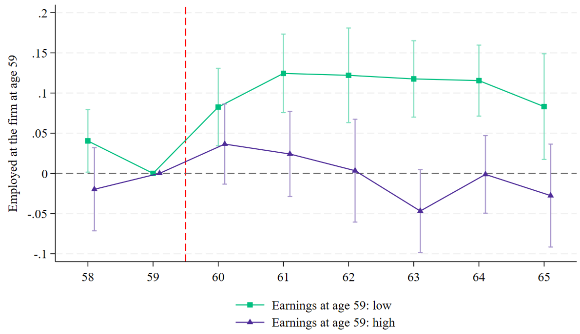
(b) From age 60 to 61–62

Notes: Figure a shows the employment effect of increasing ERB eligibility age from 60 to 61 for males. Panel b shows the employments of increasing ERB eligibility age from 60 to 61–62 for males. Employment is measured by whether individuals earn positive salary income in the following calender year they began at the stated age. The red vertical line indicates that the treatment begins at age 60. The sample is restricted to only male individuals, who were observed in the calender year they turned age 59–66, had monthly earnings of at least 88,000 JPY at age 59. The sample for younger affected cohorts in figure b is restricted to only male individuals, who were observed in the calender year they turned age 59–64, had monthly earnings of at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level. Confidence intervals shown denote 95% confidence level.

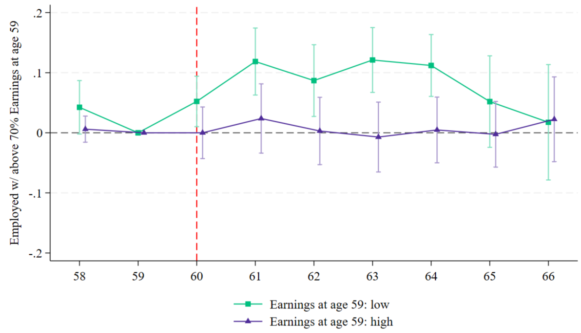
Figure 5: Heterogeneity by earnings at age 59 on different labor market outcomes



(a) Employment



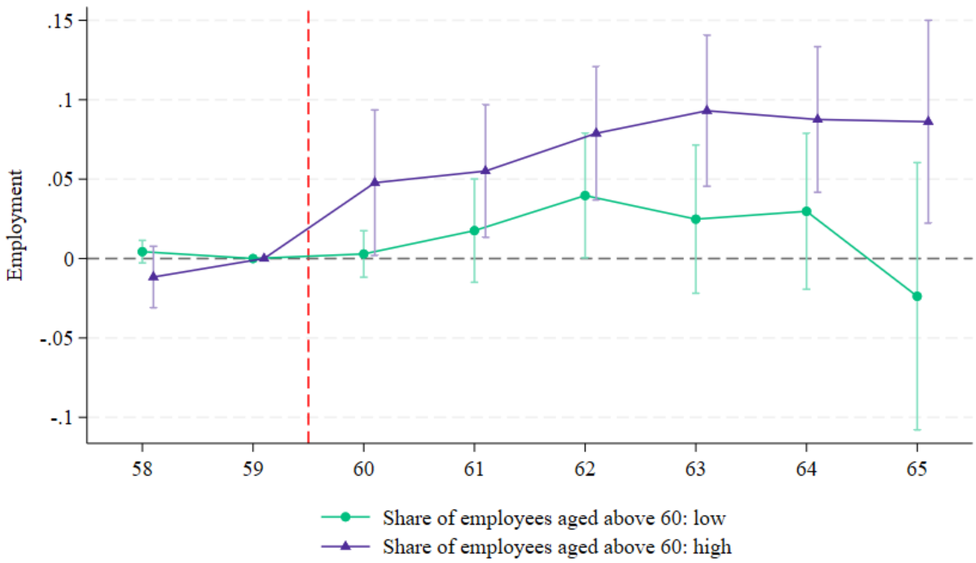
(b) Employed at same firm as at age 59



(c) Earnings above 70% of those at age 59

Notes: These figures shows the heterogeneity between high and low earners of increasing ERB eligibility age from 60 to 61 on different labor market outcomes for males. Panel a shows the effects on employment. Panel b shows the effects on the probability of being employed by the same firm as at age 59. Panel c shows the probability of being employed with earnings above 70 percent of those at age 59. Employment is measured by whether individuals earn positive salary income in the following calender year they began at the stated age. The red vertical line indicates for the age at which treatment begins. Note that this line is at 60 rather than between ages in panel c denotes that the age-60 earnings measurement is partially treated (mixing age 59 earnings with age 60 earnings within the same calendar year). Lower (higher) earners are defined by having above or below median low (high) earnings at age 59 conditional on year and municipality fixed effects. The sample for younger affected cohorts in figure b is restricted to only male individuals, who were observed in the calender year they turned age 59–64, had monthly earnings of at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level. Confidence intervals shown denote 95% confidence level.

Figure 6: Employment heterogeneity by share of colleagues aged above 60



Notes: This figure shows the employment effects of increasing ERB eligibility age from 60 to 61 for males by share of employees aged above 60 for males. The share of employees aged above 60 is measured in 2011 when all the cohorts included in this study were aged at 59 or younger. Employment is measured by whether individuals earn positive salary income in the following calender year they began at the stated age. The red vertical line indicates that the treatment begins at age 60. The sample is restricted to the three municipalities where the oldest unaffected cohorts were observed at age 59, were observed throughout ages 59–66, and had monthly earnings at age 59 at or above 88,000 JPY. Robust standard errors are clustered at the year-month of birth cohort level. Confidence intervals shown denote 95% confidence level.

A Effective Extended Years

Inspired by the "effective retirement age" posited by Rabaté et al. (2024), this section outlines the calculation of the *effective extended years* of labor supply due to the ERB eligibility age reform.

Let X_a be the random variable of the observed employment status at age a, where $X_a = 1$ denotes employment. The probability to be employed until age a for control cohort (E_a^C), is defined by

$$E_a^C = Pr(X_a = 1 \mid group = C) - Pr(X_{a+1} = 1 \mid group = C). \tag{4}$$

Similarly, the probability to be employed until age a for the treated cohort (T) is defined by

$$E_a^T = Pr(X_a = 1 \mid group = T) - Pr(X_{a+1} = 1 \mid group = T). \tag{5}$$

Therefore, the *effective extended years* (N) due to the reform during the age window of $[a_{min}, a_{max}]$ is defined by the following formula:

$$N = \sum_{a, a \in [a_{min}, a_{max}]} (E_a^T - E_a^C) \times (a - 59), \tag{6}$$

in which the $(E_a^T - E_a^C)$ indicates the change in the probability of being employed until age a due to the reform, denoted by Δ_a; $(a - 59)$ denotes the distance of age a to the one year before the pre-reform ERB eligibility age.

The change in Δ_a can be computed from the age-specific estimates for employment (β_a) from the main specification of the following models:

$$Y_{ia} = \alpha + \sum_{j, j \neq 59} D_{c(i)} \cdot \mathbf{1}\{a = j\}\beta^j + \gamma_a + \lambda_i + \epsilon_{ia}. \tag{7}$$

Hence, the change of probability to be employed until age a (Δ_a) due to the reform can be expressed as the following formula:

$$\Delta_a = E_a^T - E_a^C = \beta_a - \beta_{a+1}. \tag{8}$$

Suppose a_{max} is the retirement age when individuals fully quit the labor market, then the *effective extended year* (N) due to reform is equal to

$$
\begin{aligned}
N = & \sum_{a, a \in [a_{min}, a_{max}]} (\beta_a - \beta_{a+1}) \times (a - 59) \\
& = 0 + \ldots + 0(\beta_{59} - \beta_{60}) + 1(\beta_{60} - \beta_{61}) + 2(\beta_{61} - \beta_{62}) \\
& + 3(\beta_{62} - \beta_{63}) + 4(\beta_{63} - \beta_{64}) + 5(\beta_{64} - \beta_{65}) + 6(\beta_{65} - \beta_{66}) \\
& + \ldots + (a_{max} - 60)(\beta_{a_{max}-1} - \beta_{a_{max}}) + (a_{max} - 59)\beta_{a_{max}} \\
& = \beta_{60} + \beta_{61} + \beta_{62} + \beta_{63} + \beta_{64} + \beta_{65} + \ldots + \beta_{a_{max}} \\
& \geq \beta_{60} + \beta_{61} + \beta_{62} + \beta_{63} + \beta_{64} + \beta_{65}.
\end{aligned}
\tag{9}
$$

The first equation holds due to the assumption that there are no anticipation effects ($\beta_a = 0$ for $a \leq 59$). The last line holds under the assumption that the effects of reform are non-negative. The effective extended years of labor supply for each affected cohort is at least $\sum_{a=60}^{65} \beta_a$ years.

B Other Institutional Details in Japan

This appendix presents institutional details not covered in the main text: the details about the national pension insurance, how flat-rate benefits (FRB) and earnings-related benefits (ERB) are determined, seasonality in claiming behaviors, and other social security programs.

B.1 National Pension Insurance and Pension Amount

National Pension Insurance Insurers under the public pension scheme are classified into three groups: Type-1 insurers consist of the self-employed, agricultural workers, students, and other non-employed individuals; Type-2 insurers are those employed as salary workers, either in the public or private sectors; Type-3 insurers are spouses of Type-2 insurers with annual income under 1.3 million JPY.

Type-2 insurers, the focus of this study, are covered by the EPI. Type-1 and Type-3 insurers are covered by the National Pension Insurance (NPI). The amount of NPI benefits is the same amount as FRB for EPI insurers.

Calculation Rules for Pension FRB depends only on the duration of contribution, and the amount of ERB depends on both duration of contribution and the earnings history. Specifically, the calculation rule for the FRB is as follows:

$$FRB = 1628 \times 1.045 \times \mathcal{M} \times Duration, \tag{10}$$

where $\mathcal{M}$ is the multiplier. It is equal to one for all individuals born after April 2, 1946, which covers all individuals included in this study. *Duration* is the number of months contributed to the pension fund. It cannot exceed 480.

On the other hand, the ERB also depends on more factors as presented in the following calculation rule:

$$ERB = \left(SMR \times 0.7125\% \times Duration^{March2003}_{firstmonth}\right) + \left(SMR \times 0.5481\% \times Duration^{finalmonth}_{March2003}\right), \tag{11}$$

where SMR is the average Standard Monthly Remuneration, which is a step-function of monthly salary income. $Duration^{March2003}_{firstmonth}$ is the number of months from the time the individual starts the contribution to March 2003, and $Duration^{finalmonth}_{March2003}$ that from March 2003 to the final month of contribution.

For example, we consider a male individual born in March 1951 who earned 376,885JPY per month and had contributed 40 years. First, the corresponding monthly amount for FRB is 68,000 JPY ($\approx$ 426 USD). Second, the eligible amount for ERB is based on the number of months that individuals contributed before and after March, 2003. Specifically, using the above equation, the person could receive the ERB at age 60 at the amount of $104,960$JPY, that is $380,000 \times (\frac{52}{60} \times 0.7125\% + \frac{8}{60} \times 0.5481\%) \times 480$ months contributed., where 380,000 JPY is the corresponding standard salary remuneration to the monthly salary of 376,885. Overall, ERB accounts for approximately 60 percent of EPI benefits for EPI insurers.

For NPI insurers, the calculation rule for FRB is the same as shown above.

Claiming Rules for Those Eligibile for ERB at Age 61–64 As explained in Section 2.1, for cohorts affected by the reform of increasing the ERB eligibility age, delaying claiming ERB would not increase the amount of pension during the ages of 61–64 (Figure B1). For example, consider a male individual who was eligible for ERB at age 61 and who stopped

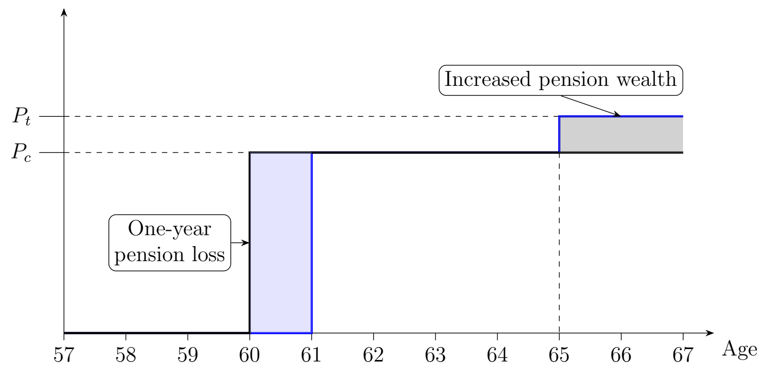


Figure B1: Effects of reform on ERB across ages

Notes: This figure shows the visual example of the change in pension wealth for an individuals who worked longer. The P_c denotes the pension level that an typical Japanese workers was eligible at age 60 without the reform. The P_t denotes the pension level that a typical Japanese worker, whose ERB was delayed by one year and who stayed one-year longer in the labor market after the reform was introduced, was eligible after he turned age 65.

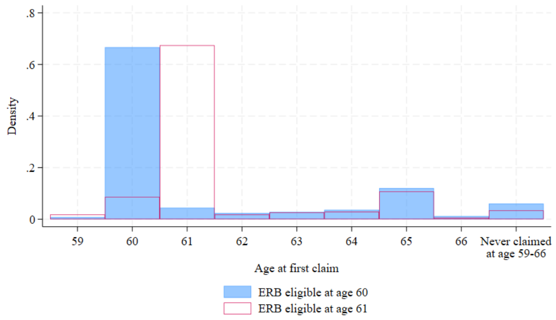
working and claimed ERB at age 61. Before the reform, he was eligible for ERB at age 60. While he lost one year of his pension at age 60 after the reform, he could start claim the same amount of ERB from age 61. The additional earnings at age 60 would increase the amount of ERB he received after he turned age 65.

B.2 Seasonality in Claiming Age

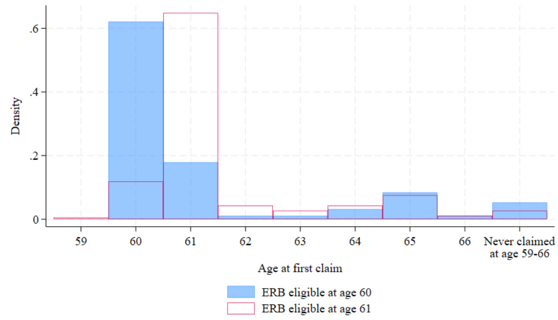
In Japan, workers may align their pension claims with the end of fiscal year, which is the March 31st in the following calender year. Evidence suggests that individuals born in the fourth quarter are more likely to delay their pension claims by one year (Figure B2). Since those born in the fourth quarter are closer to this cutoff, they may more likely to wait until the end of fiscal year, which leads to pension claiming one year after reaching ERB eligibility age. This claiming pattern is also confirmed in the raw data trend in claiming ages, as shown in Figure 2a.

B.3 Other Social Security Programs

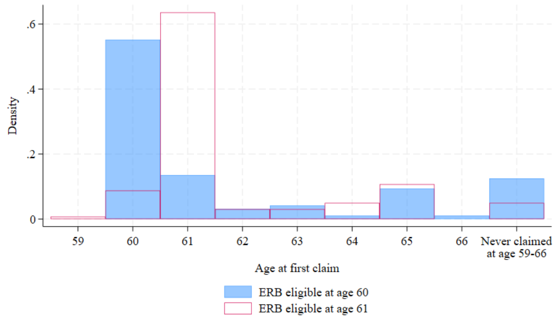
Unemployment Insurance Similar to many Western countries, Japanese workers may apply for unemployment insurance if they become unemployed before reaching pension eligibility age. The standard duration is normally one year if they are qualified, and it can be extended to two years if unemployment results from retirement at age 60 or above.



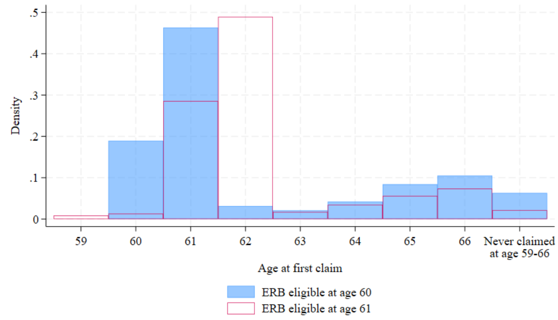
(a) Born in Quarter 1



(b) Born in Quarter 2



(c) Born in Quarter 3



(d) Born in Quarter 4

Figure B2: Distribution of ages at first claim by quarter of birth

Notes: This figure compares the distribution of the ages at first claim between individuals with ERB eligible at age 60 and 61 by their quarter of birth. Age at first claim is the age when individual first reported positive pension benefits in a given calender year. The sample is restricted to only male individuals who have available data from age 59 to 66, and had monthly earnings at or above 88,000 JPY.

Therefore, it is possible for individuals to bridge the gap to the new ERB eligibility age using the unemployment benefits for up to two years. It has also been found in other contexts that individuals rely on the Unemployment Insurance when the pension eligibility age is raised (Giesecke and Kind, 2012; Engels et al., 2017; Geyer and Welteke, 2021).

Disability Pension Individuals may also receive a disability pension if they become disabled due to illness or injury. The Basic Disability Pension is provided through the public pension system, with the amount depending on the severity of the disability. Those covered by Employees' Pension Insurance are also eligible to receive a Employees' Disability Pension in addition to the Basic Disability Pension.

Because of the strict requirements for receiving a disability pension in Japan, it is unlikely that individuals use this benefit to bridge the gap to the new eligibility age. According to the Japan Statistical Yearbook, approximately 2.1 million individuals were receiving disability pensions in 2021, representing about 1.6 percent of the population.

B.4 Institutional Backgrounds Related to Measurement Errors

Firm Pension Firm pension is provided by firms for their employees. Participation is mandatory, if a firm provides this type of pension. The age when recipients can claim this benefit is usually set to 60 or the mandatory retirement age, and recipients are allowed to defer until they retire or reach age 65.

To be precise, "firm pension" in this paper refers to both defined benefit corporate pension (*kakutei kyufu kigyo nenkin*) and EPI funds (*kosei nenkin kikin*). Most of EPI funds were transferred to defined benefit corporate pensions by 2018. Both are defined benefit pensions, for which the employee's contribution is withheld from their salary, and the difference is whether it is associated with EPI. Since we cannot distinguish them in our data, and the difference between the two systems is irrelevant to our analysis, we call both "firm pension".

Earnings Test The tax penalty due to the earnings test is determined by the sum of salary income and pre-earnings-test pension benefits. If the level of this sum exceeded 280,000 JPY per month, individuals faced an increase in the marginal tax rate by 50 percent, and if the monthly salary exceeded 470,000 JPY, the marginal tax rate increased by 100 percent. This strengthened incentives to work at older ages. However, this reform should have minimal effects on our analysis, as the cohorts we study (men born between 1952 and 1955) had already surpassed the eligibility age when the change took effect.

C Discussion for Other Potential Explanations

Focal Point Individuals may anchor their retirement decisions to salient focal ages, even when they are eligible for pension benefits earlier.

One explanation is that individuals consider the Full Retirement Age (FRA) – the age at which full pension benefits are entitled – as a focal point. In the United States, for example, many workers exit the labor force and claim benefits at the FRA of 65, despite being eligible for reduced pension from age 62 since 1962 (Peracchi and Welch, 1994; Costa, 1998; Behaghel and Blau, 2012). Similar anchoring behavior occurs when the FRA is increased to 66 Behaghel and Blau (2012). In the Japanese context, however, the mechanism is unlikely to apply. For all cohorts included in our study, the full pension eligibility is at age 65, leading to no variation across cohorts to generate differential anchoring.

Another source of anchoring behavior may come from how information is presented. Individuals may treat the "guided" or "illustrated" retirement age as the salient reference point, regardless of the actual timing of pension receipt. Evidence from randomized experiments supports this point: Brown et al. (2016) show that when the benefit schedules are illustrated using age 62, respondents report earlier expected claiming ages than when the schedule are framed around ages 66 or 70. A similar phenomenon may be relevant in our context. Age 65 may have served as a focal point due to how the increase in ERB eligibility was communicated. For instance, official guidance stated:

> In principle, (Earnings-Related) benefits can be received starting at age 65. Those who meet certain requirements may receive a *Special Old-Age Pension* before reaching 65.

The "Special Old-Age Pension" is equivalent to the ERB, but labeled differently. Moreover, official documents emphasized that this provision applied only to the affected cohorts. Taken together, these features suggest that only affected cohorts may have internalized 65 as the "principle retirement age" and adjusted their behavior accordingly.

D Appendix Tables

Table D1: Summary statistics

	(1) ERB eligible at age 60 ($e = 60$)	(2) ERB eligible at age 61 ($e = 61$)
A. Pension claims & benefits		
Has claimed by age $e - 1$	0.04	0.10
	(0.20)	(0.31)
Has claimed by age $e + 1$	0.77	0.82
	(0.42)	(0.39)
ERB at age $e + 1$ (1000JPY)	692.77	699.59
	(678.81)	(670.36)
B. Labor market outcomes		
Employed at age 59	0.99	0.98
	(0.12)	(0.14)
Employed at age 60	0.92	0.95
	(0.28)	(0.22)
Employed at age 61	0.89	0.92
	(0.32)	(0.27)
Employed at age 62	0.86	0.91
	(0.35)	(0.29)
Employed at age 63	0.82	0.87
	(0.38)	(0.33)
Employed at age 64	0.79	0.84
	(0.41)	(0.36)
Employed at age 65	0.71	0.76
	(0.46)	(0.43)
Sample	1952m1– 1953m3	1953m4– 1955m3
Observations	535	1,058
Municipalities	4	5

Notes: This table shows the summary statistics of unaffected cohorts (column 1) and affected cohorts (column 2), respectively. The sample is restricted to only individuals observed throughout ages 59–66, and had monthly earnings at age 59 at or above 88,000 JPY. Claiming pension is defined if individuals reported positive pension benefits measured in the calender year of turing that age. Employment is defined if individuals report positive earnings measured in the following calendar year of turning that age. The amount of the Earnings-related benefits (ERB) are defined at annual level.

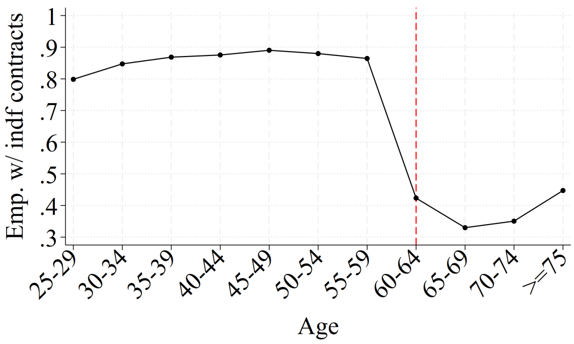
Table D2: Effects on the probability of being continuously employed

	Continuously employed between ages					
	(1)	(2)	(3)	(4)	(5)	(6)
	60	60–61	60–62	60–63	60–64	60–65
Panel A: Baseline sample						
$\mathbf{1}[ERBEA = 61]$	0.033***	0.049***	0.065***	0.060***	0.060***	0.046*
	(0.009)	(0.012)	(0.014)	(0.018)	(0.020)	(0.024)
Observations	1,593	1,593	1,593	1,593	1,593	1,593
Control mean	0.916	0.877	0.832	0.789	0.750	0.656
MOB FE	YES	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES	YES
Panel B: Lower earners at age 59 (L)						
$\mathbf{1}[ERBEA = 61]$	0.032	0.046**	0.078***	0.059**	0.058*	0.039
	(0.020)	(0.019)	(0.023)	(0.027)	(0.031)	(0.034)
Observations	797	797	797	797	797	797
Control mean	0.909	0.871	0.811	0.773	0.739	0.655
MOB FE	YES	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES	YES
Panel C: Higher earners at age 59 (H)						
$\mathbf{1}[ERBEA = 61]$	0.033**	0.053***	0.053***	0.060***	0.062**	0.056*
	(0.014)	(0.018)	(0.019)	(0.021)	(0.023)	(0.030)
Observations	796	796	796	796	796	796
Control mean	0.923	0.882	0.852	0.804	0.760	0.657
MOB FE	YES	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES	YES
H - L	0.001	0.006	-0.025	0.001	0.004	0.017
	(0.027)	(0.028)	(0.032)	(0.031)	(0.039)	(0.044)

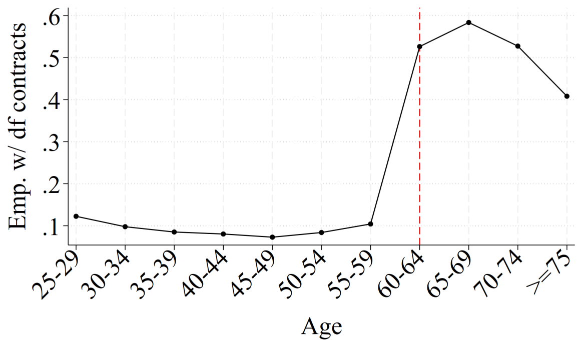
Notes: This table reports the effects of increasing the ERB eligibility age on the probability of continuously employment from age 60 until different ages. Panel a shows the effects using the main sample. Panel b shows the effects for those who had below-median earnings at age 59. Panel c shows the effects for those who had at or above-median earnings at age 59. Employment is defined if individuals reported positive earnings in the calendar year of turning certain age. The red vertical line indicates for the ERB eligibility age of the pre-reform cohorts. The main sample is restricted to only male individuals observed throughout ages 59–66, and had monthly earnings at age 59 at or above 88,000 JPY. Robust standard errors are clustered at the year-month of birth cohort level.

E Appendix Figures

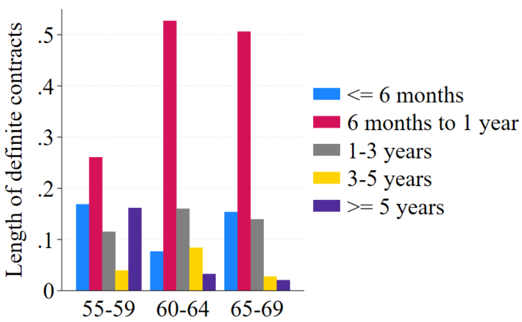
Figure E1: Contract types, length and renewals of definite contracts across ages



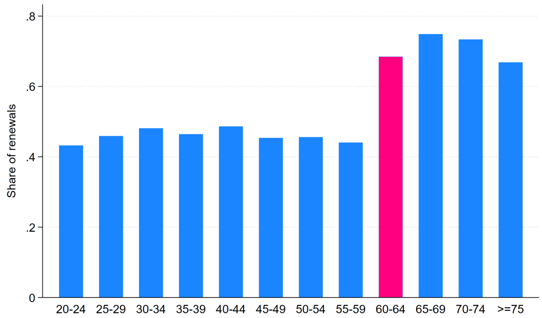
(a) Employment under indefinite contracts



(b) Employment under definite contracts



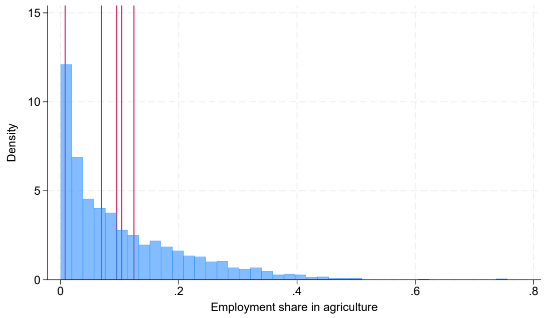
(c) Length of definite contracts



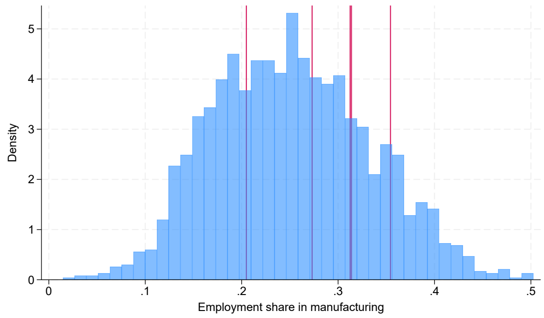
(d) Renewals of definite contracts

Notes: These graphs shows the share of employment with indefinite contracts (Panel a) or definite contracts (Panel b), the length of definite contracts (Panel c), and the probability of renewing definite contracts for non-executive workers (Panel d) by age. Indefinite contract includes "lifetime employment" contracts. Executive regular workers are normally employees at management positions. Definite contracts are normally fixed term contracts. Figures are generated by authors using Employment Status Survey (2022).

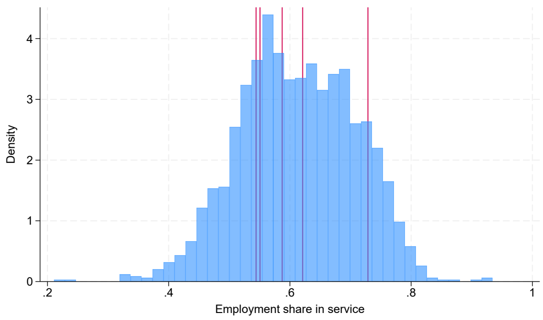
Figure E2: Distribution for employment shares and old-population shares



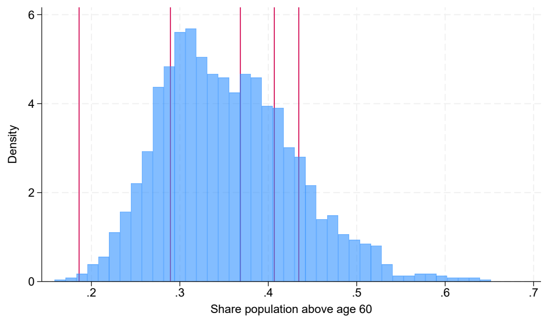
(a) Agricultural employment



(b) Manufacturing employment



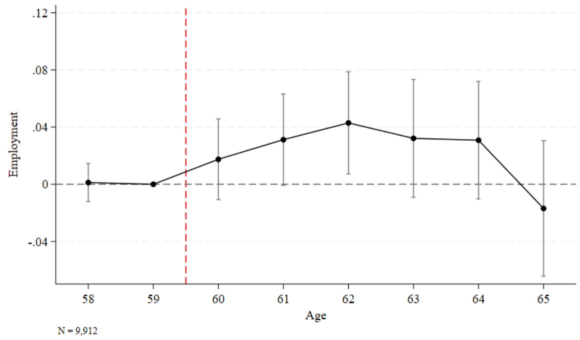
(c) Employment in service sectors



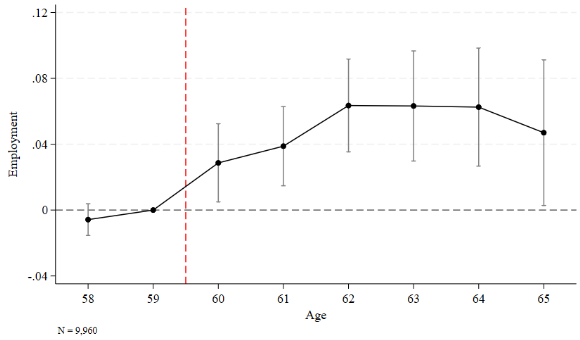
(d) Share of population aged above 60

Notes: These figures show the shares of employment in agricultural sector (Panel a), manufacturing sector (Panel b), service sector (Panel c) and share of population aged above 60 (Panel d) in 2010. The histograms show the distribution using the government official statistics in 2010 for each characteristics at municipality level. The red vertical line indicates for different municipalities included in our main sample.

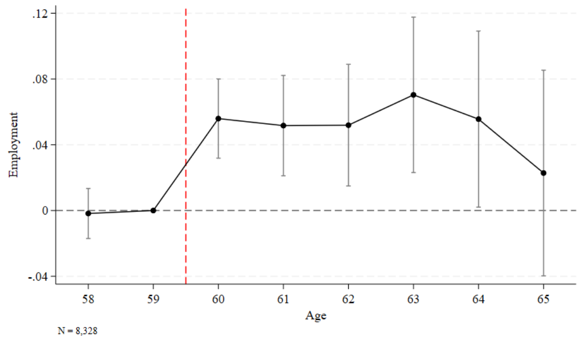
Figure E3: Robustness check for excluding different municipalities



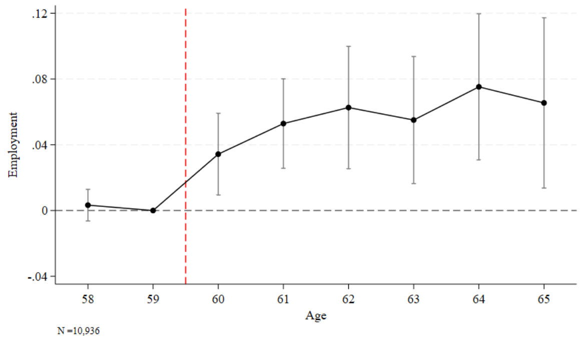
(a) Excluding Municipality 1



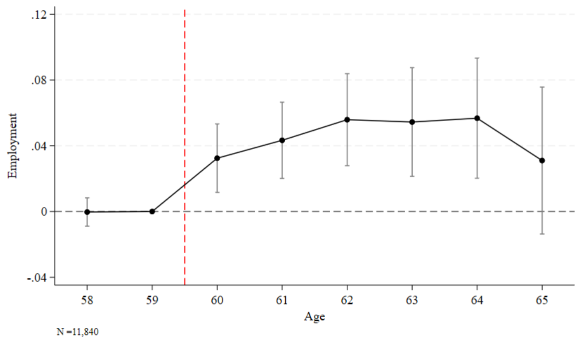
(b) Excluding Municipality 2



(c) Excluding Municipality 3



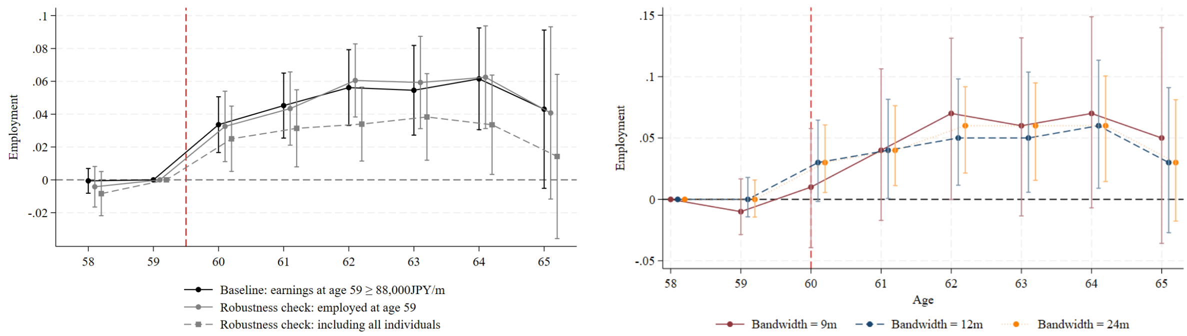
(d) Excluding Municipality 4



(e) Excluding Municipality 5

Notes: This figure shows the robustness checks for the employment effect of increasing the ERB eligibility age from age 60 to 61 by excluding different municipalities. Employment is defined if individuals reported positive earnings in the following calender year when they turn that age. The red vertical line indicates for the ERB eligibility age of the pre-reform cohorts. The main sample is restricted to only male individuals, who were observed in the calender year of turning age 59–66, and had monthly earnings at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level.

Figure E4: Robustness checks for effects of delaying ERB from 60 to 61

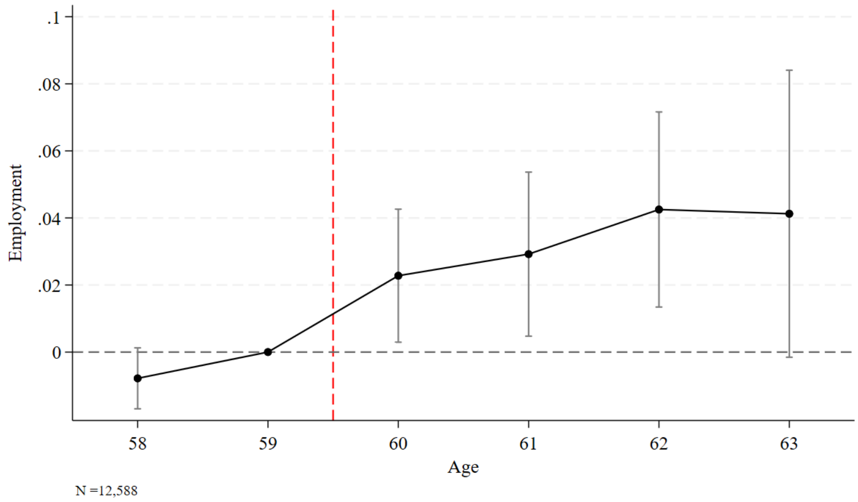


(a) Alternative samples

(b) Alternative specification

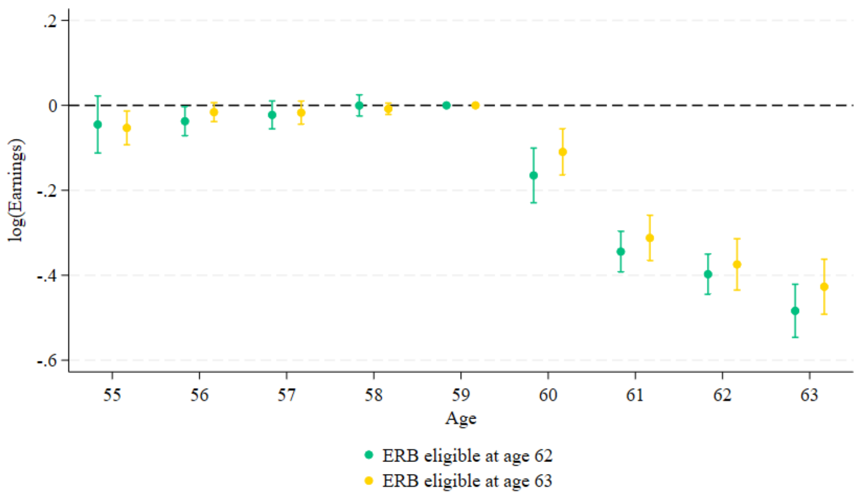
Notes: Figure a shows the point estimates and the 95 percent confidence interval of the employment effect of increasing ERB eligibility age from 60 to 61 using the main sample, and sample of individuals who earned positive earnings at age 59, and sample of individuals who earned zero or positive earnings at age 59. Figure b shows the point estimates and the 95 percent confidence interval of the employment effect of increasing ERB eligibility age from 60 to 61 using RD method across ages. Bandwidths are chosen at 9, 12, 24 months. Employment is defined if individuals reported positive earnings in the following calender year of turning certain age. The red vertical line indicates for the ERB eligibility age of the pre-reform cohorts. The main sample is restricted to only male individuals, who were observed in the calender year of turning age 59–66, and had monthly earnings at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level.

Figure E5: Effects of delaying ERB from age 60 to 62



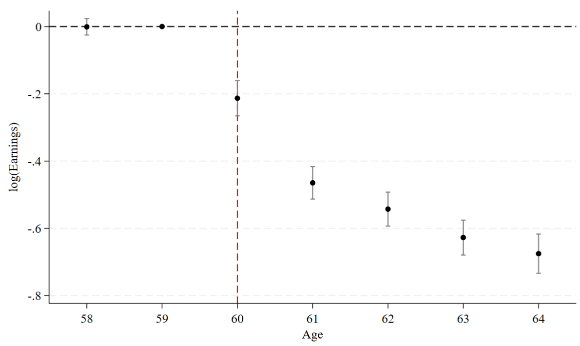
Notes: This figure shows the point estimates and the 95 percent confidence interval of the employment effect of increasing ERB eligibility age from 60 to 62 for males. Employment is defined if individuals reported positive earnings in the following calender year of turning certain age. The red vertical line indicates for the ERB eligibility age of the pre-reform cohorts. The main sample is restricted to only male individuals, who were observed in the calender year of turning age 59–66, and had monthly earnings at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level.

Figure E6: Change in annual salary: ERB delayed by two to three years

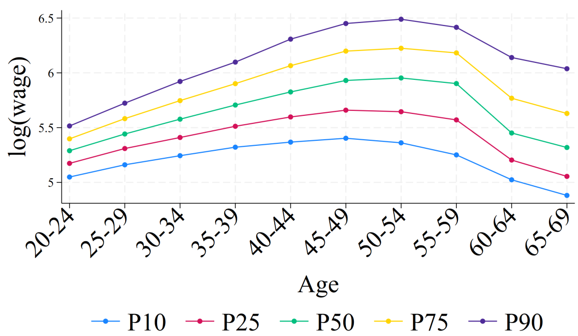


Notes: This figure shows the change in annual earnings for males with ERB eligibility age at 62 and 63 relative to the earnings at age 59 conditional on employed. The sample is restricted to only male individuals, who were observed in the calender year of turning age 59–66, and had monthly earnings at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level.

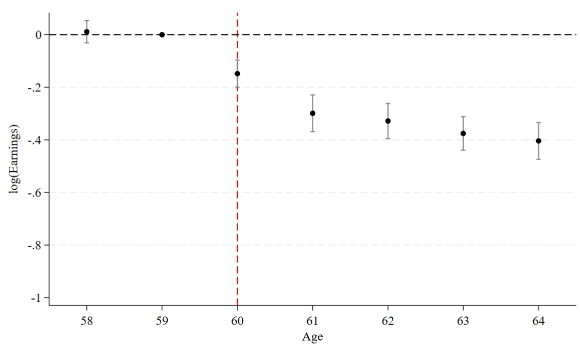
Figure E7: Change in earnings and wage across ages



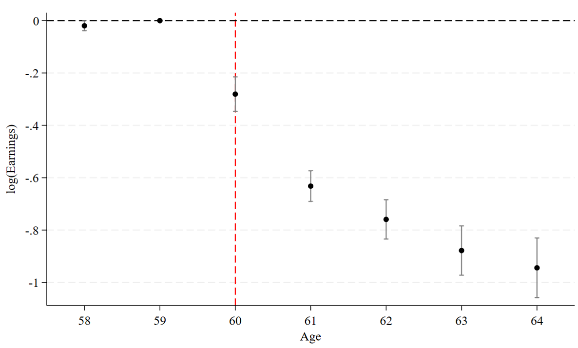
(a) Change in earnings across ages



(b) Change in wage across ages



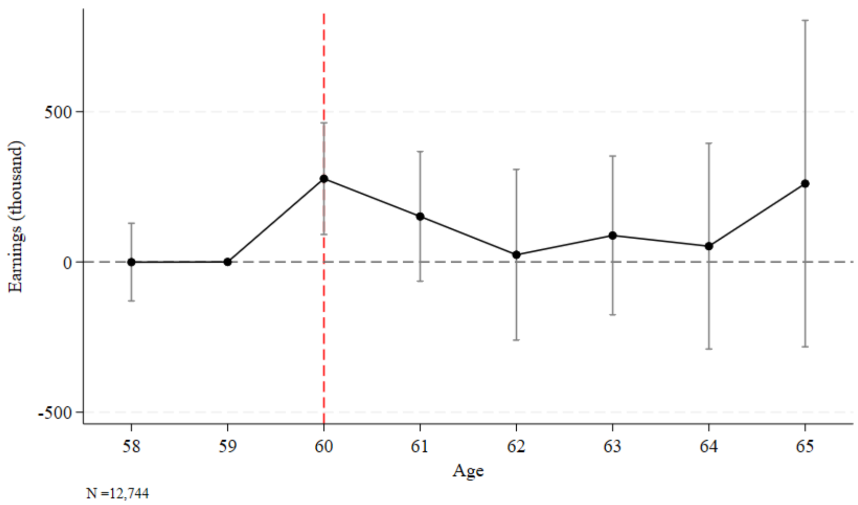
(c) Earnings at age 59: low



(d) Earnings at age 59: high

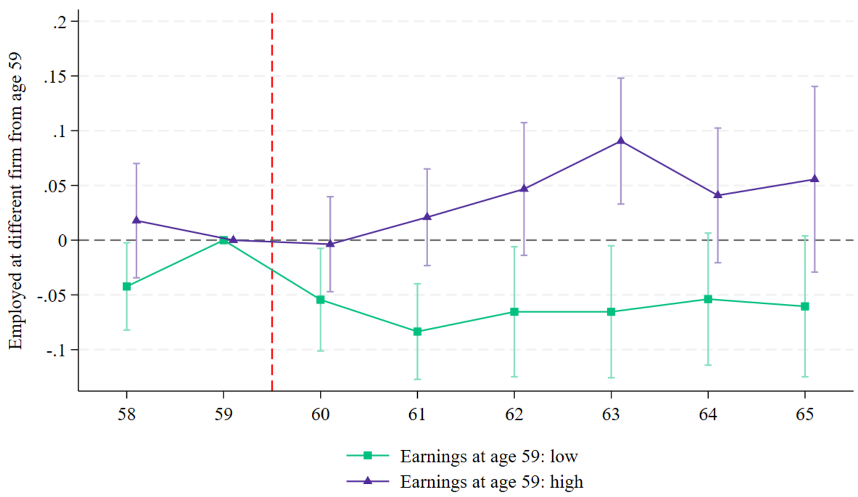
Notes: TBD

Figure E8: Effects of raising ERB eligibility age from age 60 to 61 on earnings



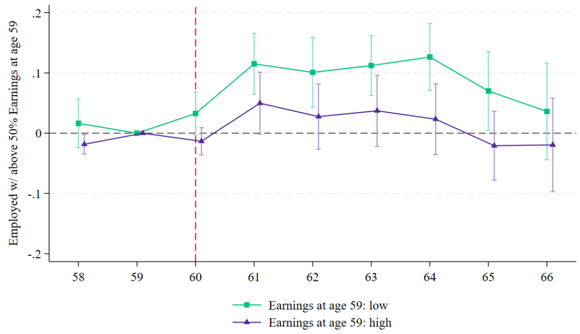
Notes: This figure shows the point estimates and the 95 percent confidence interval of the effect of increasing ERB eligibility age from 60 to 61 on annual earnings using the main sample. The red vertical line indicates for the ERB eligibility age of the pre-reform cohorts. The main sample is restricted to only male individuals observed throughout ages 59–66, and had monthly earnings at age 59 at or above 88,000 JPY. Robust standard errors are clustered at the year-month of birth cohort level.

Figure E9: Heterogeneity by earnings at age 59: employed by different firm from age 59

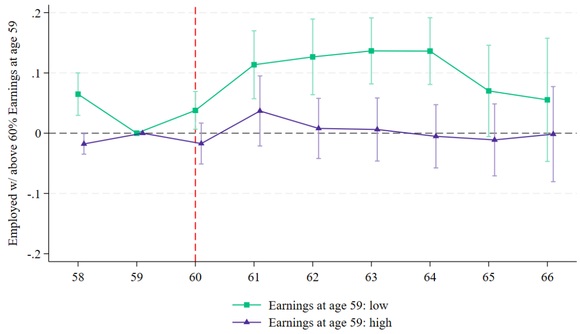


Notes: Figure a shows the point estimates and the 95 percent confidence interval of the heterogeneity effect of increasing ERB eligibility age from 60 to 61 on the probability of being employed by different firms from age 59 by earnings at age 59 using the main sample for males. Employment is defined if individuals reported positive earnings in the following calender year when they turn that age. The red vertical line indicates for the time when the effects start. Lower (higher) earners are defined if they had above-median low (high) earnings at age 59 conditional on year and municipality fixed effects. The main sample is restricted to only male individuals, who were observed in the calender year of turning age 59–66, and had monthly earnings at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level.

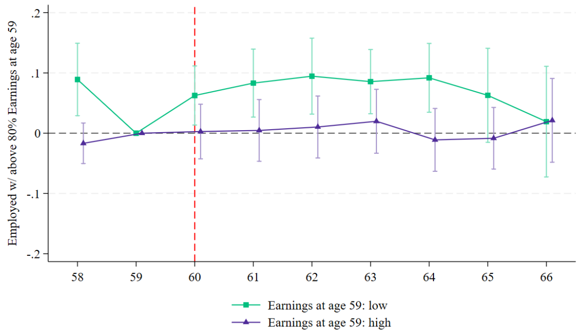
Figure E10: Heterogeneity by earnings at age 59: employed with relative earnings



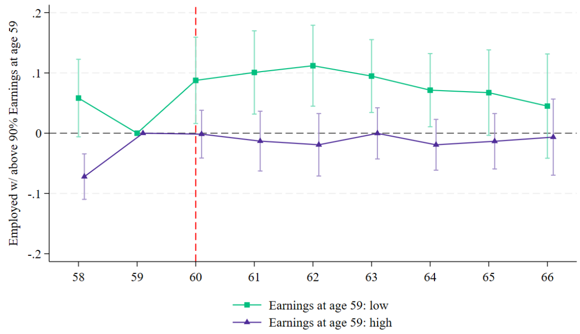
(a) More than 50% of earnings at age 59



(b) More than 60% of earnings at age 59



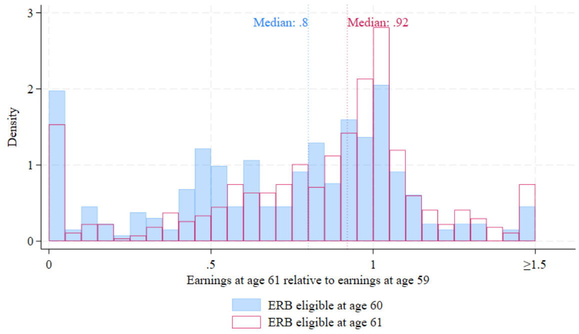
(c) More than 80% of earnings at age 59



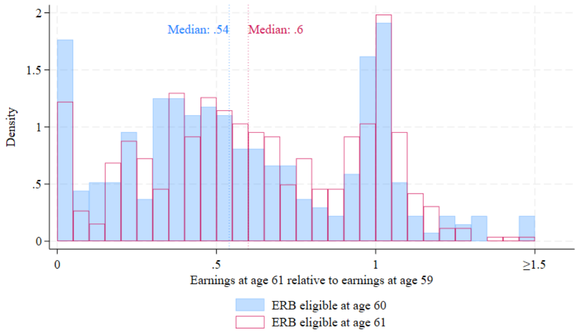
(d) More than 90% of earnings at age 59

Notes: These figures shows the point estimates and the 95 percent confidence interval of the heterogeneity effect of increasing ERB eligibility age from 60 to 61 on the probability of being employed with relative earnings to age 59 by earnings at age 59 for males. Panel a–d shows the effects on having more than 50%, 60%, 80% and 90% of earnings at age 59, respectively. The red vertical line indicates for the time when the effects start. Lower (higher) earners are defined if they had above-median low (high) earnings at age 59 conditional on year and municipality fixed effects. The main sample is restricted to only male individuals, who were observed in the calender year of turning age 59–66, and had monthly earnings at least 88,000 JPY at age 59. Robust standard errors are clustered at the year-month of birth cohort level.

Figure E11: Earnings at age 61 relative to age 59



(a) Lower earners



(b) Higher earners

Notes: This figure shows the earnings at age 61 relative to age 59 for lower earners in (Panel a) and higher earners (Panel b) using the main sample. Lower (Higher) earners are defined if they had above-median low (high) earnings at age 59 conditional on year and municipality fixed effects. The sample is restricted to only male individuals observed throughout ages 59–66, and had monthly earnings at age 59 at or above 88,000 JPY.

Figure E12: Distribution of earnings at age 61 and pension before earnings test

Notes: This figure shows the distribution of earnings at age 61 and pension before earnings test for affected males who were employed at age 60 to 61 and claimed pension at age 61. The main sample is restricted to only male individuals observed throughout ages 59–66, and had monthly earnings at age 59 at or above 88,000 JPY.